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Roles of funny HCN

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Abstract

To some extent, the main role of hyperpolarization-activated cyclic nucleotide-gated non-selective cation channels (HCN, I_h or I_f), pace-making, is dogmatized as a functional expression of one or another alpha subunit of HCN channels does not make every region of the brain or heart a pacemaker one. Recent research hints at the role of HCN in arrhythmias and seizures that are often caused by voltage-dependent K and Na channels (Kv and Nav) and neurotransmitters, respectively. There are many parallels between the HCN and K channels. Similar to Kv channels, an altered HCN function also leads to long QT interval. Moreover, a mutation in HCN is believed to trigger correlated arrhythmias and, e.g., epilepsy, among many other brain pathologies. Unlike Kv channels, although no dedicated ancillary beta subunit has been discovered for HCN, the I_h properties are also influenced by other elements and factors. A new interaction has been discovered between HCN and the vesicle-associated membrane protein (VAMP). The prevailing interaction occurs via the subtype VAMP-associated protein B (VAPB). However, this interaction is not unique but universal, since there is also a link between Kv2.1 and VAMP2 (vesicular SNARE). The most remarkable similitude is the fact that a selective antagonist of HCN and medication ivabradine prevents the I_{K_r} via the cloned human *ether-à-go-go*-related gene (HERG) channels, also known as KvLQT and Kv11.1 alpha subunit.

Keywords

HCN; I_f current; Kv2; LQT; Patch-clamp; Sag; Synaptobrevin; VAMP

...the mechanism which maintains the synchrony involves a pacemaker...

(Alexander, 1935)

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CRediT authorship contribution statement

Sodikdjon A. Kodirov: Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

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1. Introduction

Since in *Loligo forbesi* ‘we have recently succeeded in inserting microelectrodes into the giant axons of squids’ (Hodgkin and Huxley, 1939), electrophysiology and related recordings of action and resting potentials prompted the elucidation of ionic currents. Already back then, without knowledge of α subunits, one could distinguish the selectivity of channel pores based on the ions they had conducted, the direction of current flow, the reversal potential, and which antagonists were selective (Armstrong and Binstock, 1964; Frankenhaeuser, 1968; Weidmann, 1951).

There are mainly Ca, K, and Na cation-selective and non-selective ion channels (Armstrong and Bezanilla, 1974; Kodirov, 2023a; Velumian et al., 1997). Among the latter, the hyperpolarization-activated cyclic nucleotide-gated (HCN) non-selective cation channels are unique because of their simultaneous permeability to Ca, K, and Na cations. The role of HCN-mediated currents (I_f or I_h) in excitable cells was established relatively later, but the ‘very first hint for existence of these channels and their involvement in pace-making was evidenced in heart Purkinje fibers of sheep and was termed i_{K2} ’ as their kinetics were slower than other K^+ currents (DiFrancesco and Ohba, 1978; Kodirov et al., 2016; Magura et al., 1975; Noble and Tsien, 1968). Later on, an indication of i_f appeared with context of the heart, but the f here was not for nowadays ‘funny’ and HCN yet (Brown and DiFrancesco, 1980). After careful reading of one of the classical articles on *sinus venosus* tissue, I retrospectively think that the term i_f was derived from the first letter of the name of an experimental animal, the frog *Rana catesbeiana* (Brown et al., 1977). The discovery of i_{K2} and the reversal of current direction at more hyperpolarized membrane potential (E_{rev}) simply proved that the presence and function of HCN in the heart are genus- and region-dependent. The most typical HCN-mediated currents were recorded in spontaneously beating rabbit sino-atrial node tissue preparations (Brown and DiFrancesco, 1980). The phenotype of the latter currents is universal and similarly observed in neurons, e.g., in the lateral septum (LS), which are not pacemakers in contrast to those of the medial region (Kodirov et al., 2014). One may encounter the first explicit term *funny* in a study elucidating the frog atrial tissue (Goto et al., 1985). However, the I_f was masked by much bigger I_{K1} and was isolated by 100 μM Ba^{2+} ions. That is why the E_{rev} of i_{K2} in Purkinje fibers of sheep was not the same as the K equilibrium potential (DiFrancesco et al., 1979). These authors were aware that the underlying inward currents might not be only those of a pure K ion.

Depending on cells, the inward currents are carried via one of four or multiple α subunits, HCN1, HCN2, HCN3, and HCN4, which are ‘members of the voltage-gated Kv channel superfamily and cyclic nucleotide-binding domain subfamily of ion channels’ (Flynn and Zagotta, 2018). Therefore, under certain conditions, HCN may exhibit depolarization-dependent outward currents resembling those of K^+ channels that also include the presence of a tail component (Kodirov, 2020; Kodirov et al., 2024). Also, there are resemblances in the functioning of pore and voltage-sensitive domains (PD and VSD) among the latter types of channels. Moreover, the auxiliary subunit of K^+ channels, MinK-related peptide 1 (MiRP1), slows the activation time constants of HCN4 from ~400 to 700 ms (Decher et al., 2003). Biophysical changes occur via the C-termini of KCNE2 and HCN2 or HCN4 channels. This interaction was selective, as it was absent with regard to KCNE1, 3, and

4 subunits. Another parallel is the modulation of HCN and Kv channels by the pituitary adenylate cyclase-activating polypeptide (PACAP) signaling pathway (Johnson et al., 2019).

Similar to Kv channels, the HCN is also influenced by other proteins. As a novel observation reveals (David et al., 2019; Silbernagel et al., 2018), the I_h could be increased by the vesicle-associated membrane protein (VAMP). VAMP is a constituent of the soluble N-ethylmaleimide-sensitive factor attachment protein receptor complex (SNARE) and is important for the secretory release of neurotransmitters (Bark, 2009). Although there are several associated with HCN elements, this is the second functional one along the tetratricopeptide repeat-containing Rab8b-interacting protein (TRIP8b) present in *cornu ammonis* area 1 neurons of the hippocampus (Brager et al., 2013). The VAMP-associated protein B (VAPB) did not alter any other properties of HCN2 in oocytes except for amplitude when activated during steps up to excessive -180 mV from a holding potential (HP) of -30 mV, and independent of whether the configuration was either on-cell or inside-out (Silbernagel et al., 2018). VAPB exhibited selectivity toward HCN1 and HCN2 compared to HCN4, KCNQ1, Kir2.1, Kv1.1, Kv1.4, and Kv4.3 channels. The subtype VAPA was less effective than VAPB, while VAPC was ineffective on HCN2. The combination of VAPA and VAPB additively increased the I_h via HCN2.

As a side effect, the glucagon-like peptide type 1 (GLP-1) accelerates the cardiac frequency in female landrace pigs, and ivabradine did not influence it, though the GLP-1 effects were targeted to SAN (Lubberding et al., 2024). It can be concluded that either the GLP-1 is more potent in modulating the pacemaker rate or the modulation occurs independent of HCN channels. Nonetheless, the effects of GLP-1 perhaps occur indirectly via other ion channels during the repolarization. For example, the Cav agonist S-Bay K8644 decelerates the rate of contractions in hiPSC-CM cultures (Lickiss et al., 2024), a property that is prevalently assigned to funny channels.

The designated modulators of HCN channels are considered to be cAMP and cGMP. However, as it can be expected, the latter two have more targets, and 100 nM 8-Br-cGMP facilitates the inhibitory effects of S-nitrosoglutathione (SNAP) on spontaneously beating isolated hearts from *Achatina fulica* (White et al., 2004). Thus, the negative chronotropic effects of this NO donor may occur via cGMP, albeit irreversible. The concentration of cAMP in sensory neurons of *Aplysia californica* is elevated by serotonin, and this effect is paralleled by inhibition of K channels and prolongation of spikes (Abrams et al., 1984).

Although many mutations can be achieved experimentally for HCN and some play a role in cAMP binding of the channel, required for proper activation, their impact at the organ level is not conclusive. For HCN4 channels, based on cAMP-binding, the C662, E660, L652, R669, R710, R713, and T670 mutations are defined (Wojciechowski et al., 2024). Among them, only the C662 did not bind 8-Fluo-cAMP. Even binding sites exhibit not a strict selectivity, and therefore, any of these mutations in isolation without crucial impact for severe pathologies or even physiology of organ.

The overlap in properties of HCN and Kv channels extends to their potential roles in pathologies, e.g., epilepsy and LQT syndrome (Powell et al., 2015; TeBay et al.,

2022). Designated blocker and medication ivabradine antagonizes the outward currents in HEK293 cells expressing HERG, the human *ether-à-go-go*-related gene channels, also known as Kv11.1 (Fiore et al., 2025). Since moderate blockade of Kv channels by class III antiarrhythmic agents decelerates the repolarization phase and thereby prolongs the APD and QT interval, the same can be expected with the ivabradine. Excess blockade of HERG channels may lead to triggered early after depolarization (EAD) during cardiac AP. Unique alternating EAD in field potentials is triggered in human induced pluripotent stem cell-derived cardiomyocytes (hiPSC-CM) by an anticancer aminopyrimidine inhibitor, nilotinib, which is potentially an off-target effect on HERG (Izumi-Nakaseko et al., 2025).

Arrhythmias are also observed in the Genetic Absence Epilepsy Rat from Strasbourg (GAERS) *in vivo*, though the QT intervals and heart rate were not significantly altered in contrast to *in vitro* (Powell et al., 2015). Perhaps the HCN channels are a trigger for the absence seizure since the expression of HCN2 and HCN4 subunits was slightly lower when assessed at the mRNA levels in GAERS with documented signs of epilepsy at adulthood. Ivabradine decreases the HCN-mediated sag in the cortex and ventrobasal nucleus of the GAERS model and perhaps thereby prevents seizures to some extent (Iacone et al., 2021). However, oral ivabradine alone was ineffective, and absence seizure numbers decreased only during concurrent with elacridar exposure. The role of HCN is also explored in a mouse model of Alzheimer's disease (La Barbera et al., 2021). However, when the density of channels in dopaminergic neurons of Tg2576 mice was assessed, an observed decrease in amplitude was not confirmed.

Therefore, the main gap in the literature is the lack of in-depth analyses of raw data. By highlighting the actual experiments, this review will focus on the impact of HCN channel activities and underlying sag in membrane potential (MP) on the physiology and pathologies of cells, including those during concurrent arrhythmia and epilepsy.

2. Tale of *funny* current

It all started with a heart and Purkinje fiber of a sheep from 'Oxford Co-operative Society Slaughterhouse' (Noble and Tsien, 1968), in which, when the extracellular solution contained 2 mM KCl, upon hyperpolarization to -88 and -105 mV from HP of -65 mV, a gradually activating inward current i_{K2} resembling that of I_f (+HCN) was registered, while at -129 mV, an inactivation with distinctively faster kinetics was observed (Fig. 1A). Retrospectively, it can be assumed that the latter was of Kir origin (+ I_{K1}). Interestingly, at -113 mV, once activated, no further gradual changes occurred in the current waveforms. However, upon termination of these four steps, comparable deactivation (+Tail) of hyperpolarization-activated currents occurred in a voltage-dependent manner and most plausibly were of the same HCN origin as in *figure 5* (Noble and Tsien, 1968). Note that 'when cited in italics as above and elsewhere, the reader is prompted to learn from the corresponding figures and tables in the referred studies. Also note that each study will be analyzed extensively, so information until the next citation belongs to one study' (Kodirov, 2023b). Additionally, a citation indicating the original authors of figures will be provided immediately next to it.

The absence of I_f at -129 mV ($-HCN$) could be because of domination by I_{K1} in Fig. 1A, and the underlying funny currents may be unmasked (dotted theoretical trace) as in frog atrial preparations (Goto et al., 1985). Unless it is unique to sheep Purkinje fibers, the absence of I_{K2} at -113 mV (considered a reversal potential) could plausibly mean that inward currents via both I_f and I_{K1} are canceling each other because they perhaps were of the same amplitude and kinetics, albeit the polarity of relaxation. However, this notion is too theoretical and requires thorough modeling with a deep knowledge of the cell membrane and the two types of ion channels as a whole unit. The very first typical (for all kinds of cardiomyocytes and neurons) inward currents were recorded in the SAN of rabbits and were designated as i_f (Brown et al., 1979). Note that the same figure appears identically slightly later (Brown and DiFrancesco, 1980), since the former article apparently was planned to be of a review type but was published earlier for some reasons.

Before the aforementioned ion channel conductance was termed i_f it was just an ‘additional current’ with respect to i_{K1} (Brown et al., 1977). The ‘additional current’ in turn was formerly known as i_{K2} and later also as i_X or $i_{X,slow}$ depending on genus and heart regions (Brown and DiFrancesco, 1980). Therefore, in some subsequent studies, the corresponding conductance was still referred to as a novel hyperpolarization-activated K^+ current (Araque and Buño, 1994), as it mediated the anomalous rectification in MP responses - sag. Although, the latter authors were aware of I_f , I_h , and I_q , as the same conductance is termed in different cells (Kodirov et al., 2014, 2016).

One of the insightful recordings was obtained from the sinus venous of *Rana cateabeiana* (Brown et al., 1977). The authors have first recorded spontaneous pacemaker action potentials (AP). Thereafter, they have just switched to a voltage-clamp mode slightly before and after maximum diastolic potential (MDP). These experiments are highlighted in figure 4 (Brown et al., 1977). Logically, during the former switching, the amplitude of the additional current was much higher. Nonetheless, the kinetics of currents activated at two physiologically different times had a resemblance. It is because the ‘records have been photographically superimposed’ or, in other words, the amplitude or the start of additional current was not considered. Retrospectively, it is obvious that the second smaller current develops much slower. This study is further insightful since the sinus venous of *R. cateabeiana* had TTX-sensitive and insensitive pacemaker regions based on a concentration of $6 \mu M$. The latter exhibited the I_{si} , a slow inward current. Next, in the sinus venous tissue of *R. cateabeiana*, a comparison was made with regard to both combined current/voltage-clamp and plain recordings in a voltage-clamp mode (Fig. 2A and B), respectively. The HP was set to -46 mV since the MDP of the recorded tissue had this particular value. In contrast to the combined approach (Fig. 2A), the hyperpolarization to -51 mV in voltage-clamp mode did not activate the additional current ($-HCN$), and consistently, no tail was observed (Fig. 2B). Upon the -56 and -61 mV steps, a clear-cut, gradually developing current was registered along the tails. Under these conditions, the kinetics of tails were different.

In general, the electrophysiological recordings are robust and insightful, and one can assess the function of multiple types of ion channels at the same time in a single native cardiomyocyte or neuron. Even in some cells, two or more types of α subunits among the Cav, HCN, Kv, Kir, and Nav channels can be simultaneously activated under

identical conditions with the same voltage protocol (Kodirov, 2023a). It is both logical and physiological since during a single AP, several ion channels are activated, deactivated, and inactivated sequentially.

One of such experiments revealed sequential activation of I_K and i_f in rabbit SAN tissue (Brown and DiFrancesco, 1980). Depolarization of MP by a prepulse to -20 mV from a HP of -40 mV activated I_K , and a subsequent repolarization to -50 mV triggered i_f that gradually developed during 500 ms steps (Fig. 3A). Upon the return to HP of -40 mV an inward tail of i_f was generated. Upon repolarization to -50 mV it is possible that i_f was contaminated by the deactivation of I_K since the tail was outward based on holding current at -40 mV. Therefore, when the prepulse to -20 mV was absent, the amplitude of i_f was smaller, and the entire current was inward upon repolarization to -50 mV. However, the contamination did not significantly influence the deactivation of i_f since the contributing tails were of comparable amplitudes. With the prepulse, the amplitude of i_f was bigger since the cell membrane was depolarized by I_K , and thereby the repolarization was functionally intense. When the repolarization was achieved at -60 mV, there was no contamination, and the amplitude of the i_f and tails was accordingly increased (Fig. 3B). However, note that under both conditions, the amplitudes of tails are very similar, despite the bigger i_f after the prepulse. Upon hyperpolarization to -70 mV, the tail amplitudes were even bigger but again similar (Fig. 3C). Under these conditions, the prepulse to -20 mV had a lesser effect on the amplitude of i_f , with a smaller difference at the start of the hyperpolarization step of -70 mV. Note that recordings were fairly stable, and during the entire course of experiments, the instantaneous currents upon the prepulse to -20 mV have only slightly increased (Fig. 3A–B).

3. HCN and pacemaking

High-frequency recurrent APs are attributed to inward currents termed funny I_f despite the ‘simple fact that HCN channels are present throughout the brain and heart, but prevailing only the medial septum in the former and SA/AV nodes in the latter exhibit a pacemaker electrical activity’ (DiFrancesco et al., 1991; Kodirov, 2015). The SAN dictates the physiological heart rate (HR). Therefore, cardiomyocytes in SAN are true pacemakers compared to any excitable cells in general and to atrioventricular (AV) ones in particular. There is a parallel between abnormalities in HR and epilepsy, along with mutations in HCN (Powell et al., 2015). Some epilepsy-related mutations are designated as highly ‘likely pathogenic HCN1 variants’, including C329S and V414M among G391C, G391D, G391S, I397L, K261E, M153I, M243R, M305L, R590Q, and S399P (Marini et al., 2018). Other parallels are also described for the involvement of identical agents and receptors in the modulation of cognition and heart function (Tamano et al., 2018). Many substances, including the classical neurotransmitter serotonin at a 30 μ M concentration, may modulate HCN (Gasparini and DiFrancesco, 1999).

For potential translational purposes, often human embryonic stem cells are studied (Ormrod and Ehler, 2023), which may also differentiate into cardiomyocytes. For some reason, cultured cardiomyocytes or stem cells of various origins (also, those from bone marrow; Kodirov, observation) exhibit unexpected pacemaker-type spontaneous activities. However,

the electrical property of induced pluripotent stem cell-derived cardiomyocytes (iPSC-CMs) was concluded to reflect ventricular cardiomyocytes but not SAN cells (Silbernagel et al., 2018). Although the results with regard to HCN are robust in mice and zebrafish, it is not clear why the same channels were not targeted in mouse iPSC-CMs, but instead Cav and Nav1 were studied. Indeed, based on the AP phenotype, the iPSC-CM is also considered ventricle-like by others (Pahlavan et al., 2018), but their unstable RMP is not taken into account. In control iPSC-CM, the contraction rate was 59 per min⁻¹, which decreased to 48, namely by 18 % in those with the long QT (LQT) phenotype. The LQT syndrome is of particular interest, but there is not always a clear-cut genetic mutation linked to it (Donahue, 2018).

A decrease in iPSC-CM contraction did not correlate with the degree of QT interval prolongation of 37 % (Pahlavan et al., 2018). Also, the AP initiation and termination with the maximum diastolic potential (MDP) resemble those of pacemaker cells. The presence of MDP reveals that iPSC-CMs are prone to pacemaker activity and are similar to other stem cells (Bett et al., 2016). MDP is obliterated by 150 nM zatebradine, which is an HCN antagonist (Izumi-Nakaseko et al., 2017). The zatebradine decreases the frequency, accompanied by a prolongation of action potential duration (APD), which is consistent with its side effects on Kv1.5 channels in iPSC-CM. Since the tested concentration of zatebradine is lower than the EC₅₀ of 480 nM for SAN cells, it additionally substantiates that the iPSC-CM is rather reminiscent of pacemakers than the working myocardium (Kodirov, submitted).

4. HCN and sag

The activation of HCN currents (I_h) and sag in MP often correlate when channels are sequentially activated in neurons during voltage- and current-clamp recordings (Kodirov et al., 2014; Solomon and Nerbonne, 1993). The presence and recording of sag can be traced back to pioneering studies using the giant axon of *Loligo*. As responses of MP to step hyperpolarization were complex, upon its termination, the RMP was not restituted immediately but after a ‘heavily damped oscillation’ (Hodgkin and Huxley, 1952). Unlike action potential (AP), the sag results from the gradual activation of HCN alone.

The aforementioned properties of neurons and the presence of sag in lamina II/III of the cortex and the lateral amygdala (LA) of adult rats are exemplified in this context (Fig. 4). The sag amplitude, e.g., in adult cortical neurons, is -6.1 mV, which is consistently accompanied by a 5.5 mV rebound tail potential (RTP). The sag and RTP (Fig. 4A), or their magnitude, are interlinked; if an agonist or antagonist is effective, then it influences both parameters (Lehnhoff et al., 2019; Solomon and Nerbonne, 1993). The absence or presence of sag could be used in order to distinguish distinct firing types of adult neurons (Fig. 4A and D). The sag and RTP are present in superior colliculus-projecting (SCP) neurons of the primary visual cortex (Solomon and Nerbonne, 1993). Moreover, the occurrence of sag and RTP correlates with the presence of I_h and tail currents in SCP neurons.

In order to elucidate the I_h currents, the neurons isolated from P3 rats were transfected with either WT or L157V HCN1 or a combination of two (Bonzanni et al., 2018). The

rationale of this and other articles is that up to P10, the contribution of endogenous I_h to MP or inward currents is insufficient. But the immature neurons between P0 and P9 were not studied. It is impossible that the ~5 mV sag suddenly appears at P10 both in the dendrite and soma of LV pyramidal neurons of the neocortex as in *figure 4* (Atkinson and Williams, 2009). This study does not even mention the sag since the focus was on the somato-dendritic voltage attenuation, i.e., during simultaneous recordings at P10, the response of the dendrite to hyperpolarization of MP was only ~20 % lower vs. the soma, while at P27 it was 90 %. Responses were evoked by ~1 nA current injection, which is usually excessive for immature neurons.

After transfection of P3 rat neurons with the WT and/or L157V HCN1, the expected expression possibilities were termed hetero- in contrast to homomere channels (Bonzanni et al., 2018). However, the actual heteromerization should be proven by other methods than electrophysiology, especially under these conditions and for this mutant. What are the chances that every neuron will express heteromeric channels, and why should the total amounts differ for current- vs. voltage-clamp experiments, namely 0.5 vs. 1.5 μg ? In this case, will a functional correlation in terms of channel properties and excitability occur adequately?

Although the expression of sag is robust in some types of cells (Barnes and Hille, 1989), it is not always adequately resolvable upon mild hyperpolarization in neurons (Fig. 6A). Moreover, occasionally the counterpart of sag that occurs as inward currents in the same neurons can be activated only by high-magnitude steps (Fig. 6B and C). This uncorrelated activity in voltage-clamp mode may occur because of HP values of -70 and -50 mV, which are intermediate compared to RMP values of -61.1 mV in vitro.

The correlated presence of I_h and sag in MP is sometimes accepted with skepticism. The same skeptical view applies also to the simultaneous presence of sag and rebound action or tail potentials (RAP or RTP). These views can be substantiated by the low ~1 pA/pF density of I_h at -140 mV in true pacemaker cells (Hassinen et al., 2017). At P28, a significant sag was enabled by ~300 pA steps in the cholinergic interneuron of the striatum (Sugasawa et al., 2018). However, this near-maximal current amplitude led to a very mild hyperpolarization of -90 mV. The neuron was spontaneously active at baseline, but RAP most probably resulted from the termination of this hyperpolarization. Under these conditions, 4 % sevoflurane, which is used to anesthetize animals, obliterated spontaneous AP and RAP, but considerable sag and RTP were still present compared to the effects of the selective HCN antagonist ZD7288 on the P21 neuron in *figure 5* (Sugasawa et al., 2018). In these cells, there is an increase in input resistance (R_{in}) in the presence of sevoflurane and ZD7288. The peak amplitude of I_h was ~300 pA at -130 mV, and both drugs were less effective compared to the outcomes of current-clamp experiments. Even in the presence of 40 μM ZD7288, the inward tail was still present, which was activated at -60 mV. Note that the tail, if solely linked to HCN activation, is sensitive to 1 μM ZD7288 and obliterated by 10 μM (Kodirov et al., 2014). Despite its prevailing selectivity toward I_h and tail current, the ZD7288 also blocks Cav channels at overlapping concentrations (Sánchez-Alonso et al., 2008). In this case, the IC_{50} was 40.9 μM , and the inhibition did not reach 100 % but only 77 % even at a 1 mM concentration of ZD7288. In sheep cardiac Purkinje fibers, the effective

concentration (EC_{50}) of ZD7288 was lower in inhibiting I_f activated at less hyperpolarized potentials, i.e., at -100 mV, it was 850 nM (Berger et al., 1994). The latter is consistent with the EC_{50} of 920 nM when the MDP was assayed. At -140 mV, >3 μ M ZD7288 was required for I_f inhibition. Interestingly, the amplitude of I_f in Purkinje fibers is extremely high at 42 nA at -140 mV, though the same mean value was obtained at -120 mV. In cells of a true pacemaker - SAN of guinea pigs, the amplitude of I_f is only ~ 500 pA but highly sensitive to ZD7288, and effects are observed even at 300 nM (BoSmith et al., 1993).

Changes in sag and other cell properties are rapid during the early postnatal days (Fig. 4). When the ages of male mice were significantly different, spanning from postnatal day 7 (P7) to P35, the neuronal morphology and size also differed (Sugasawa et al., 2018). However, differences in excitability and sag amplitude cannot be conclusive when $n = 1$ for P28 or $n = 2$ for P7 neurons. A clear-cut HCN-mediated sag is often observed at more negative MP (Fig. 6A), but in some preparations, it is robust and is triggered even by a -50 pA step (Evans and Dougherty, 2018). Use-dependent Nav1 antagonist carbamazepine depolarized the RMP in the dorsal hippocampus (DHC) by 3 mV but did not alter the sag in MP. The presence of HCN in DHC is also substantiated by subsequent occurrences of RTP. The RMP and sag remained unaltered in the ventral hippocampus (VHC). The membrane time constant (τ_m) was significantly increased in VHC vs. DHC, though this is perhaps the impact of one out of six neurons in *figure 7* (Evans and Dougherty, 2018). Comparable changes in R_{in} of VHC were not considered significant. Either R_{in} or τ_m was estimated at -65 mV, but the actual RMP, which apparently was more depolarized, is not explicitly stated. The τ_m was estimated from voltage responses to 1 ms $\cdot 100$ pA steps. Since ± 0 and $+100$ pA steps are not absolute because of constantly injected currents to artificially hold the MP at -65 mV, it could bias the comparisons among DHC and VHC neurons.

Neurons may tolerate the injection of excessive currents of positive polarity (Fig. 4E and F), but often not a negative one, e.g., -350 pA (Figs. 4C and 10C). In addition, note that the input resistances (R_{in}) were ~ 0.4 , 2.3 , and 1.5 G Ω in WT HCN1, untransfected, and L157V HCN1 neurons, respectively (Bonzanni et al., 2018). The R_{in} at G Ω level is definitely high, and such neurons, whether newborn or adult, are incapable of handling -350 pA, at least not for seconds (Figs. 4C and 5D). In the case of the combinatory expression of L157V and WT HCN1, the R_{in} was 1.9 G Ω . Note that the R_{in} is high since these cortical neurons were immature as they were isolated from P3 rats and kept in culture for up to 3 DIV - days in vitro. When SCP neurons of the primary visual cortex were isolated at P4 and kept in culture for 14 DIV, the R_{in} was still ~ 0.9 G Ω in spite of heterogeneity, and a significant hyperpolarization was achieved just by -70 pA currents (Solomon and Nerbonne, 1993). Even in adult cells, an injection of -300 pA occasionally leads to instability in the MP response, and therefore -280 pA is used (Kodirov et al., 2014). The best physiological approach is not to irrelevantly hyperpolarize the MP of cardiomyocytes and neurons further than -100 mV. In this regard, the importance of R_{in} could be proven when perfect patch parameters in whole-cell mode change significantly over time. An increase in access resistance (R_a) and R_{in} occurs often during the course of long experiments, and the latter may alter in response to certain agents that block voltage-dependent channels. Under any of the above mentioned circumstances, an injection of high-amplitude currents, even around -200 pA, will be negatively consequential (Fig. 4C). However, sometimes this type

of instability could be temporary, and upon subsequent milder steps, the MP may again respond adequately, e.g., already at -160 pA (Fig. 4C). Besides, based on the MP response (Bonzanni et al., 2018) and the Ohm's triangle, the R_{in} is ~ 0.1 M Ω (Fig. 10C). Note that sometimes the amplitude of sag may reach the plateau earlier than 1 s (Fig. 4A, C, and D vs. Fig. 6A). This might correlate with the presence and differential activation kinetics of certain HCN subunits.

Since the experimental MP, set to -70 mV, and true RMP values are apart by up to -30 mV, perhaps the authors were able to inject a -350 pA current (Bonzanni et al., 2018). Thus, the magnitude of an actual current step might be lower (Fig. 10C). However, the response of MP and appearance of sag are not typical (Figs. 4A, 6A, and 7B), although they exhibit some similarity to pyramidal neurons of lamina V in the anterior cingulate cortex (ACC) of P56 male mice (Santello et al., 2017). In ACC, the sag and RTP have comparable phenotypes, whether recorded from the dendrite or perikaryon. Usually, in immature and certain types of neurons with high R_{in} , the MP requires marginal currents in order to respond either with sufficient hyperpolarization or firing (Solomon and Nerbonne, 1993). An immature phenotype of exogenous progenitors is retained even after migration to a certain lamina of the cerebral cortex when recorded in adult mice (Muzzi et al., 2007). In one such neuron, when recorded in the mouse cortex, a 5-pA current step already triggers repetitive spikes, as revealed in Fig. 5D. This is a universal phenomenon: the active and passive properties of neurons depend upon age (if postnatal) and stage (if embryonic). However, the vast majority of studies emphasize body mass, especially those subjected to behavioral paradigms. Postnatal life span dependency on drugs or paradigm effects is plausible for brain regions (Hill et al., 2019). The universality equally concerns all regions of the brain, and when adults are tested, the neurons will have much lower R_{in} (Kodirov et al., 2021).

5. HCN and RMP

The involvement of HCN in RMP is not expected in cardiomyocytes and neurons, but the values of MDP in SAN could substantiate it in pacemaker cells if the I_f is fully or partially inhibited. Even the most adequate modulator of HCN, cAMP at 1 mM extracellular concentration, could depolarize the RMP of thalamocortical neurons only by ~ 5 mV (Budde et al., 2005). Therefore, studying and comparing any effects based on RMP with context to HCN is difficult. Although, occasionally, with the help of a mathematical approach, a perfect difference of 1 mV between -66.3 ± 0.2 vs. -67.3 ± 0.2 mV for $n = 4$ vs. 3 in the respective ACI and WAG/Rij rat groups has been documented. The ACI refers to August x Copenhagen-Irish rats serving as a control, while WAG/Rij denotes the Wistar Albino Glaxo ones bred in Rijswijk, which are considered a genetic model of absence epilepsy (Kodirov, 2023b).

A potential role of HCN alone in pacemaking or RMP is not considered enough; therefore, its interaction with other channels is assumed. The latter is possible, but the notion that HCN are 'key channels modulating the resting membrane potential (RMP) in thalamocortical neurons' or any other excitable cells is without merit (Rivero-Echeto et al., 2021). The HCN is permeable for several cations, and how should they obey the Nernst law while

simultaneously contributing to the RMP of the neuron? Even if the I_f contributes to RMP, then what contribution have the I_{K1} (Sato, 2002) and other K channels?

Effects on spike frequency, R_{in} , or RMP are also mediated by many types of channels. In an adult cortex, the instantaneous frequency of spikes was 18.6 Hz, which exhibited irregular adaptation (Fig. 4A). At the end of the 1-s pulse, the rate was 12.2 Hz, as measured after the injection of 200 pA of current. The average frequency during a 1-s pulse was 14.6 Hz. Since the adaptation index was low and there was no double spiking at the beginning of the AP train, this cell is not a fast spiking (FS) interneuron, as shown, e.g., in the barrel cortex (Geoffroy et al., 2013). The instantaneous frequency in the lateral amygdala neuron was calculated between the 1st and 2nd spikes (Fig. 4F), which resembled double spiking at 50.6 Hz upon the 300-pA step. In this neuron, using a sparse increment, a rheobase for current was determined at +100 pA. Although the frequency of total action potentials ($n = 296$) has steadily increased, the somatic responses to sequential steps are not strictly gradual. Note that ‘nervous messages are invariably associated with an electrical change known as the action potential’ (Hodgkin and Huxley, 1939).

The G protein-activated inwardly rectifying K^+ channels (GIRK) maintain the RMP of excitable cells, and alteration of this parameter will influence the properties of other channels. For example, cocaine inhibits homomeric GIRK2 and heteromeric GIRK1/2 (Kobayashi et al., 2011). In locus coeruleus (LC) neurons, cocaine decreases the frequency of spikes by triggering regular oscillations, which in turn enable the bursting behavior, and regular spikes are generated during up-states (Williams and Marshall, 1987). The number of spikes is also constant at 2 and is controlled by the duration of each oscillation, since the 3rd AP is terminated even at threshold (Fig. 11).

The RMP shift by up to 10 mV after caffeine, cocaine, and a combination of two in *figure 2A* (Rivero-Echeto et al., 2021) will not substantiate the role of HCN. First, the exact values are required, rather than the ‘approximate RMP’. Second, there are no additive effects derived from both stimulants. Third, more depolarized RMP was observed during acute caffeine + cocaine, which slightly increased the I_f at more hyperpolarized MP. Fourth, both agents dramatically decreased the I_f after a chronic state, and the corresponding behavior of RMP is not stated, though an opposite and much more significant shift is expected based on *figure 6A* (Rivero-Echeto et al., 2021). Fifth, the caffeine + cocaine is obliterating instantaneous currents at -50 and -60 mV from a holding potential of -40 mV, which is not a hyperpolarization and thus the underlying channels are not HCN ($\downarrow$, Fig. 13A and B). It is not hyperpolarization, as the RMP of these neurons is not -40 mV. Sixth, the authors are concluding the effects of binge on both HCN and Cav3, with the possibility that there is an interaction between these remotely related types of channels. Finally, identical traces are represented as controls for the acute and chronic groups of caffeine + cocaine binge (Fig. 14). The latter is perhaps achieved by low-pass filtering traces from the acute group in *figure 1B* based on lower capacitance currents in chronically treated mice in *figure 6A* (Rivero-Echeto et al., 2021). Also, it is not clear why and how ‘average current traces’ were obtained for $n = 6$. These recordings are described as obtained in P25 mouse brain slices.

In trout sinoatrial node (SAN) cells, the I_h was present, though its phenotype was not robust even during 4-s pulses (Hassinen et al., 2017). The amplitude of total inward currents was small (100 pA) and consisted of a 50-pA pure I_h component. The pacemaker APs possessed an MDP of about -60 mV, which is more depolarized compared to the RMP values of atrial and ventricular cardiomyocytes. In contrast, the atrial and ventricular cardiomyocytes did not exhibit I_h under identical conditions. However, the expression of HCN3 and HCN4 was not selective for SAN, although it was higher compared to the atrium and ventricle. Comparable tendencies were observed for HCN1, HCN2, and KCNE2 channels. Similar to mammals, the HCN3 of fish possesses the cyclic nucleotide binding domain (CNBD). The HCN4 CNBD is also comparable to that of humans, albeit with distinct C-termini. All four HCN α subunits were also present in the brain of the fish *Salmo trutta fario*. It substantiates the notion that HCN is a determinant but not the sole player leading to pacemaker activity and RMP (Kodirov, 2022).

Hence, which type of channel is prevailingly contributing to RMP in VB neurons? In *figure 3B* (Rivero-Echeto et al., 2021), the values are from the depolarizing test steps but not 'holding pulse potential', since the HP was -110 mV. Again, in *figure 3A, B* (Rivero-Echeto et al., 2021), the 'membrane potential suddenly reached its resting value at the end of the inhibitory input' is inadequate, as currents are via ion, but not receptor channels. Consequently, this is not neurotransmission. Thus, it cannot favor the generation of large or small low-threshold spikes (LTS). In *figure 7C* (Rivero-Echeto et al., 2021), the timing lasting 50 ms prior to the ramp is not sufficient to activate HCN channels, and sag is not substantiated by this type of protocol. The term LT AP with regard to the generated spikes upon the threshold during ramp is erroneous based on a height of ~100 mV. Besides, the MP was not held at -65 mV as stated above, but at -70 mV for all groups. If the membrane potential was held at -65 mV, how much current was injected, and how did this affect the magnitude of the steps?

In non-pacemaker cells, it is plausible that HCN, concurrently with Kv channels, may modulate the RMP. Concurrent heterologous expression of KCNE2 with HCN4, but not HCN3, enables I_h in CHO cells (Hassinen et al., 2017). Interestingly, HCN3 alone was also not able to produce any currents. The 84 % of I_h in SAN was blocked by 2 mM Cs⁺, while the AP frequency was blocked by 19 %. These non-correlated effects could be explained by the fact that 2 mM Cs⁺ also blocked the I_{Kr} by 63 %. Subtracted 21 % effects on both currents hint that the blockade of the latter channel is compensated for by that of the former. However, this is more of a hypothesis, as the I_h was of low amplitude and revealed in only 4 out of 16 cells, despite the presence of HCN3 and HCN4 transcripts in all. This is consistent with previous observations on the endogenous HCN3 α subunit in HEK 293 cells (Varghese et al., 2006). The absence of currents in CHO cells is perhaps related to the low expression levels of HCN3. Although the current waveforms resemble each other, the amplitude of HCN3-mediated ones is smaller compared to the HCN2 channel (Mistrić et al., 2005). Surprisingly, the half-maximal activation of HCN2 and HCN3 occurred at an identical -95 ± 1 mV.

Introducing α HCN2 into the *Xenopus* oocytes depolarizes the RMP to -32 vs. -43 mV in spite of heterogeneity in results (Pittoors and Van Bogaert, 2015). This particular modulation

of RMP by HCN2 may involve an interaction with the I_{K1} channel. The concurrent functioning has also been documented for the I_h and muscarine-sensitive outward K currents I_M in motoneurons (Buskila et al., 2019). However, this synergy does not alter the RMP. The substance P reversibly increases the holding current for ~90 s at -60 mV in DRG neurons (Ishimatsu, 1994), but I_h was not sensitive in contrast to I_M , hinting that the HCN role for RMP is minimal. Note that the magnitude of the holding current often robustly predicts the value of RMP in excitable cells (Kodirov, observation).

The RMP of native neurons was depolarized by transfection with WT HCN1 to about -40 mV, while in the case of mutated ones, it remained unaltered at -51 mV (Bonzanni et al., 2018). How to explain the outcome that the combination of WT and L157V HCN1 only slightly hyperpolarized the RMP to -53 mV? In the case of current densities, it is plausible that the presence of WT HCN1 is ineffective as the L157V mutant suppresses the channel function in a dominant negative fashion. The latter feature is also similar to those of Kv channels. Differences of ~10 mV in RMP in the absence or presence of exogenous HCN1 are solid (Bonzanni et al., 2018) compared to another recent study. By mainly addressing the HCN and sag in the ACC, the RMP values of -66.16 vs. -64.96 mV in the WT and LP-211 groups were documented (Santello et al., 2017). Despite the acceptance in the field and consistency with the literature, how could a Δ of -1.2 mV be considered an effect in vitro and have a true impact in vivo? Secondly, instead, our unpaired (since only mean values are available) *t*-test reveals a *P* value of 0.5 along with a standard deviation (SD) of 2.9 vs. 2.7 mV. Even a Δ of 0.05 mV for the sag ratio was found to be significant, in contrast to the recalculated *P* = 0.08 as above. Is the R_{in} really that low? Considering Ohm's law, a simple arithmetic yield of 0.03 M Ω after dividing ~15 mV of hyperpolarization by -400 pA of injected currents. Thus, it cannot be ruled out that the 'coupling of 5-HT₇Rs to HCN channels is independent from the pathological state' (Santello et al., 2017). Finally, as already stated, there was no difference in MP waveforms, i.e., the sag and RTP, when recorded from apical dendrites vs. soma.

Although KCl is present at high concentrations in the intracellular milieu, K⁺ gluconate is used for patch-clamp experiments as it enables stable and long-lasting recordings. The latter salt is also used for the comparison of HCN-mediated sag in MP (Kodirov et al., 2014). However, K⁺ gluconate may alter the properties of HCN channels without affecting the RMP (Velumian et al., 1997). The most precise approach, i.e., recording RMP of pyramidal neurons of the hippocampus first in KMeSO₄ and then replacing it with K⁺ gluconate within the patch pipettes, revealed respective average values of -62 vs. -61 mV. That is a Δ of 1 mV, and the stated mean difference of 0.1 mV might be a discrepancy. Either way, it contradicts the notion that HCN may modulate the RMP since K⁺ gluconate was concluded to reduce I_h along other major inward and outward currents. Although not analyzed, why does K⁺ gluconate increase the sag in *figure 1* (Velumian et al., 1997)? The authors correctly state that an increase in R_{in} accompanied the decrease in I_h by K⁺ gluconate. The effect of K⁺ gluconate was reversible, and an increase in sag proportionally altered the RTP. Among agents targeting the HCN, zinc could also be of interest because of its reversible potency (Araque and Buño, 1994). Similar small effects are observed with other compounds (Kodirov et al., 2014). Therefore, one considers pure I_h amplitude, which is an inwardly rectifying current that gradually increases immediately after the onset of the hyperpolarizing

step (Fig. 7C and D) and reaches the steady-state level within seconds, depending on brain regions and the prevailing presence of certain HCN α subunit. The rate of activation is also voltage-dependent and is faster upon stronger hyperpolarization. Saturation in amplitude after a period of time reflects a maximum activation of HCN channels at given conditions.

Sometimes, under voltage-clamp conditions, the activation of HCN channels is observed only at strongly hyperpolarized potentials, which is not attributed to distinct values of the holding potential (Fig. 6B, C, and F). In contrast, the response of MP to the current protocol in the same neuron is typical, and prominent sag is recorded at wider ranges (Figs. 6A and 7B). The sag amplitude increases according to the magnitude of hyperpolarization. In this LS neuron, however, the amplitude of RTP was very small despite the presence of significant depolarizing sag. The underlying I_h was activated first at a holding potential of -70 mV (Fig. 6B) and, subsequently, at -50 mV (Fig. 6C) in the same neuron. The comparison and subtraction of traces (Fig. 6D) recorded under both conditions revealed no differences, despite the increased magnitude of I_h at a holding potential of -50 mV. The obvious activation of HCN channels and the slow equilibration of currents were observed only at -140 mV (Fig. 7E) and were independent of the tested holding potentials of -70 mV (○) and -50 mV (●). At a HP of -50 mV, the amplitude of the tail currents remained unaltered, while the instantaneous current magnitudes were similarly increased at the tested potentials (Fig. 6C). The patch quality at these two holding potentials was almost identical, which is evident at least upon hyperpolarization to -140 mV (arrows) since no capacitance artifacts in the subtracted traces were present (Fig. 6C and D). However, upon depolarization back to a holding potential of -50 mV, the artifacts of RAP - action currents - were observed, as clearly emerged in subtracted traces (*) and shown in an expanded time scale (Fig. 6H). Also, the amplitudes of AC were truncated in Fig. 6D and now appear in full scale. Thus, the I_h in this LS neuron possessed a distinct property that was not derived from the value of holding potentials. The lack of slow equilibration by HCN channels at less hyperpolarized steps (Fig. 6B and C) contradicts the observed I_h -mediated sag at a broader range of MP under current-clamp conditions (Fig. 6A). Nonetheless, the activation of HCN channels, depending on each subunit, may still take place if prolonged (>4 s) hyperpolarization is used (Altomare et al., 2001). However, it is not clear that when “each subunit comprises a voltage sensor moving between ‘reluctant’ and ‘willing’ states, whereas voltage sensors are independently gated by voltage, channel closed/open transitions occur allosterically” (Altomare et al., 2001). Also, the tested MP range of up to -300 mV as shown in *figure 9* (Altomare et al., 2001) is not physiologic either for the willing or reluctant distribution parameter (z) or open probability – P_o dependency. This biophysical allosteric model prediction was based on *fine-guess* model parameters for HCN2 α subunit. Note that there are no P_o values at -50 mV, and that of z starts barely at this voltage. Interestingly, no word *funny* appears in this article by discoverers (Altomare et al., 2001), although any study addressing the HCN uses it or at least mentions the pacemaker channel.

Finally, although it is evidenced that the selective blocker of I_h ZD7288 decreases the amplitude of instantaneous currents at almost all tested pulses (Kodirov et al., 2014), only the latter antagonist may substantiate how big the contribution of HCN to both inward currents and thereby to RMP is. Especially the concentration-dependent inhibition of the I_h tail is one of the most reliable indices (Kodirov, 2020). However, note that there are

two components of I_h , comprising voltage-dependent and independent currents (Proenza and Yellen, 2006).

6. HCN and Kv channel parallels

Based on several molecular features, these two types of cation channels have similarities and therefore belong to the same channel subfamily with a cyclic nucleotide-binding domain (Flynn and Zagotta, 2018). Nevertheless, the HCN and Kv channels are unique since the underlying currents have a clear-cut phenotype. Both types of channels are crucial to the generation of action and membrane potentials in the brain and heart.

The expression of HCN channels in brain regions has been demonstrated indirectly using the patch-clamp technique (Kodirov et al., 2014; Solomon and Nerbonne, 1993). Depending on the relative expression levels of four α subunits or their biophysical properties, the I_h parameters may vary (Budde et al., 2005). For example, a permanent open state for HCN2 is proposed (Pittoors and Van Bogaert, 2015).

It has been concluded that the HCN channels in the prefrontal cortex (PFC) 'are key regulators of synaptic integration in prefrontal pyramidal neurons but *that their* functional contribution is dependent on a *partnership* with Kir2 and K_{leak} channels' and 'this inference was corroborated by experiment' (Day et al., 2005). However, the conclusion is solely based on EPSP summation during the computer simulation in *figure 8* (Day et al., 2005). It is not clear why the outcomes of the simulation are not tested in real neurons and experiments, especially when the HCN, Kir2, and K_{leak} currents were recorded and simulated in parallel. Nonetheless, the expression of HCN was insightfully delineated. In single pyramidal neurons, the HCN1 and HCN2 were expressed at mRNA levels, while in PFC tissue additionally the HCN3 and HCN4. The neurons were from prelimbic and infralimbic (PrL/IL) areas, and therefore the 'pooled PFC' blot in *figure 2* (Day et al., 2005) is perhaps also restricted to these regions. However, *figure 3* (Day et al., 2005) is less insightful because despite the presence of 1 μ M TTX, the deactivation of HCN in A is not as gradual as in B or C, and it is possible that in the former the action current was triggered upon return to HP of -50 mV from -120 mV. Therefore, the artifact of AP (see Fig. 6C and H) should appear in subtracted ZD7288-sensitive currents in *figure 3B* (Day et al., 2005). Besides, no currents during the wash-in of ZD7288 are presented. In *figure 3C* (Day et al., 2005) perhaps also subtracted traces are depicted, since they appear similar to those in B, but it is not stated explicitly.

HCN2 is also expressed in the perikaryon and terminals of cortical neurons, and exogenous VAPB similarly targets these regions (Silbernagel et al., 2018). A consistent observation was made when HeLa cells were used after introducing both exogenous HCN2 and VAPB. However, co-expression of the mRNAs of HCN2, VAPA, and VAPB is not convincing in brain sections and cannot unravel the ineffectiveness of VAPA. No obvious selectivity toward the brain or spinal cord areas is seen. The two interneurons in the internal granular cell layer of the cerebellum in *figure 4D* (Silbernagel et al., 2018) cannot be appreciated at this magnification, and the presence of additional cells belonging to that type is expected.

The HCN current density in thalamic neurons of VAPB^{-/-} mice decreased to 15 vs. 22 pA/pF in WT (Silbernagel et al., 2018). The latter changes did not alter the activation parameters. Since the activation of I_h requires longer pulses during milder steps, an adequate dynamic protocol was used. An excessive hyperpolarization was avoided at more negative test pulses by considerably shortening the duration of steps gradually to ~3.75 s (see Fig. 18). The most negative MP was set to -130 mV, which could be considered the physiological limit of any excitable cell. It would be even more insightful if the sag in MP of VAPB^{-/-} neurons was elucidated in parallel with I_h (see Fig. 6). As the MP and thereby the bioelectricity influence not only the molecular elements and physiology but also the anatomy of the organ (Pietak and Levin, 2018). Specifically, the epileptiform activity (EA) experienced by each neuron may alter the profile of somatic elements in a complex manner, as, e.g., treatment with kainite (KA) increases the somatostatin receptors in both the amygdala and hippocampus at mRNA levels, but related seizures decrease them in the former region only (Iwasawa et al., 2019). The increased expression in the amygdala was in turn targeted primarily at pyramidal neurons. This could hint at the specific role of the amygdala as the prime target for corticotropin-releasing hormone-triggered seizure (Baram et al., 1992). The latter is observed in EEG in the form of occasional paroxysmal discharges in the cortex and hippocampus in P5 to P22 rats. The CNS neurons of the invertebrate are also prone to epilepsy-related paroxysmal depolarization (Altrup and Speckmann, 1988; Brenes, 2022).

The association of HCN with other channels is not unique, whether functional or physical. Also, the link between either VAPA or VAPB, even with Kv2.1, is not unique since a third or fourth member, the N-terminal domain-interacting receptor 2 (Nir2) or Nir3, 'specifically associate' with them (Kirmiz et al., 2019). How is it specific when there are two receptors, the Nir2 and Nir3, if not considering both VAMP proteins? The authors term the association of the channel with VAPA or VAPB as Kv2.1 complexes. Most experiments are conducted in HEK293T cells, and in native cells there are also Kv2.2 subunits.

The neuronal VAMP2 (vesicular SNARE) binds to the N-terminus of the Kv2.1 channel (Lvov et al., 2009). VAMP2 may affect the interaction between the N- and C-termini. In oocytes, coexpression of exogenous VAMP2 with Kv2.1 shifts the half-maximal inactivation voltage ($V_{0.5}$) to the left. Although no traces are shown, the shift is approximately -5 mV in the simultaneous presence of VAMP2, whereas Δ NVAMP2 had opposite effects, although not statistically significant. It was concluded that the N-terminal domain (NTD) of VAMP2 is the major player in modulating the Kv2.1 channel property. Kv2.1 consistently shows U-type inactivation, which is more pronounced in the presence of VAMP2. Interestingly, VAMP2 still shifted $V_{0.5}$ to the left when the N-terminus of Kv2.1 was truncated (Lvov et al., 2009). Thus, the N-terminus was not the determining factor. However, the experiments are not precise because in *figure 3A* (Lvov et al., 2009), the $V_{0.5}$ was approximately -15 vs. -20 mV, and in *figure 2C* (Lvov et al., 2009), the corresponding values change to -20 vs. -27 mV in the presence of VAMP2. The control experiments should be more consistent and reproducible, but this is the reality of the average data.

VAP is a transmembrane protein and is expressed in the endoplasmic reticulum. VAP binding sites on sequences are referred to as FFAT-like motifs, which are also present in

the Kv2.1 and Kv2.2 channels (Furuita et al., 2021). Because of conserved residues, VAPA interacts directly with both channels. In this case, the affinity of the interactions is highest for Kv2.1 vs. Kv2.2, as well as among the other tested peptides.

A vesicle-bound protein VAMP2 is also known as synaptobrevin and was elucidated in squid (*Loligo pealei*) giant synapse preparations of the stellate ganglion (Llinas et al., 1994). Presynaptic and postsynaptic responses were recorded with sharp microelectrodes, and the effects of tetanus toxin (TeTX) on spikes were compared. Presynaptic injection of TeTX at a concentration of 19 μM did not change the parameters of postsynaptic spikes after 35 min. This period was sufficient for the diffusion of TeTX into the terminal, which was monitored by the fluorescence of the fluorophore and was completed after 18 min. The rate of diffusion was relatively slow because of the narrower tip of the electrodes during conventional electrophysiology. In contrast, the L-chain of TeTX at a concentration of 26 μM gradually abolishes the postsynaptic spike. Both variants of TeTX have no effect on presynaptic spikes. Under these conditions, the amplitude of AHP is higher in presynaptic spikes. In the presence of the L-chain of TeTX, the amplitude of the AHP in postsynaptic spikes increases significantly after 18 min, despite a small decrease in AP height. At 34 min, the amplitude of the AHP continues to increase while the amplitude of the AP remains the same. At 37 min, no spike is elicited, but a neurotransmitter-mediated depolarization (EPSP, excitatory postsynaptic potential) occurs. After 46 min, the L-chain TeTX decreases the amplitude of EPSP. Although the time points in the two experiments are not identical, the effect on the AHP is unique and proves that a whole TeTX has no effect. In this case, a small difference in concentrations (19 vs. 26 μM) may not be critical. However, the above observations are indirect, as they were confirmed by the cleavage of synaptobrevin with TeTX in the squid optic lobe (Llinas et al., 1994). The amount of synaptobrevin was similar in both homogenates and synaptosomes. TeTX decreased synaptobrevin comparably in synaptosomes of the squid optic lobe and the rat brain. The molecular weight of synaptobrevin is different in these species.

7. HCN and epilepsy

The relative and region-dependent contribution of HCN channels to CNS neuronal physiology is evident (Figs. 4, 6–8). Therefore, it is logical to expect their participation in pathologies, including epilepsy. Causes for epilepsy via HCN are rare compared to K^+ and Na^+ selective channels, but perhaps more frequent than, e.g., the pyridoxine-dependent seizure occurring in 1 out of 700,000 livebirths (Perucca and Perucca, 2019). The same is true when one compares HCN-caused epilepsy with those triggered by neurotransmitter receptor channels (Niquet et al., 2016). Among 597 patients with epilepsy, only 17 HCN-related heterozygous missense variants were identified (DiFrancesco et al., 2019). Except for 2 out of 17 mutations, these were single cases, and thus, a total of 19 patients, or 3.1 %, had HCN genetic abnormalities. The association of HCN with APBA2, Caveolin-3 (CAV3), filamin A (FLNA), GRASP, KCNE2, MAGI2, or PEX5L did not always reveal pathology.

A mutated HCN1 has been found to lead to genetic generalized epilepsy (GGE) in 1 out of 136 patients (Bonzanni et al., 2018). Note that the presence of HCN and pacemaker activity is associated with spontaneous high-frequency AP firing, but the L157V HCN1

mutant was concluded to further accelerate it as the channel exhibited decreased I_h currents. When the active properties of neurons were compared, the untransfected ones exhibited regular spiking in response to 40 pA steps, while in the presence of WT HCN1, the excitability was prevented. In contrast, the absence of HCN4 in SAN cells slows the rate of spontaneous APs (Baruscotti et al., 2011). Besides, in neurons, the influence of HCN channels on depolarization-evoked spikes is considered absent (Koch and Grothe, 2003). However, the APs are influenced by 5 μ M ZD7288, though the effects in *figure 1* (Koch and Grothe, 2003) are complex. 1) It is inhibiting sag and blocking rebound action potential in onset firing neuron, but some of the HCN channels are still active and contributing to RTP; 2) the sag and RAP are blocked also in ‘adapting’ neuron (though it is not a typical gradual adaptation), but it is enabling additional bursts; 3) it blocks the sag in a regular spiking (RS) neuron termed ‘sustained’ and even triggering a burst upon the onset of depolarization. Nevertheless, the rheobase current that is around +300 pA for the aforementioned experiments and the three types of neurons is high, since some WT HCN1-expressing neurons are excitable upon 50 pA steps (Bonzanni et al., 2018). But the average rheobase current was +138 pA, which is also high based on the evoked spikelets in *figure 6* (Bonzanni et al., 2018) and MP depolarization close to ± 0 mV. Another difference in the presence of WT HCN1 is the generation of AHP upon the termination of the pulse in *figure 5A* (Bonzanni et al., 2018). Cells were excitable at overlapping ~ 20 Hz when the expressions of L157V alone or concurrently with WT HCN1 were achieved. Mutations of C-terminal F486, L530, and V628 did not prevent VAPB-mediated increases in I_h via HCN2, while the truncation of the N-terminus did (Silbernagel et al., 2018). Also, the replacement of the N-terminus of HCN2 by that of HCN4 made VAPB ineffective.

8. Epilepsy and heart rate

Animal models of epilepsy demonstrate that pathologies occur not only in one area of the brain or a single organ (Kodirov, 2023b; Powell et al., 2015; Powell and O’Brien, 2015). Seizures in the brain and an alteration in HR may occur concurrently in vivo (Mamalyga, 2014).

However, convincing experimental data is hard to obtain in vitro. In this regard, some relatively influential studies will be highlighted in greater detail within the scope of this assay. One of the represented experiments concerns the concurrent occurrences of absence epilepsy and sinus dysrhythmia in mice lacking the pacemaker channel HCN2 (Ludwig et al., 2003). Also, from introductory and theoretical points of view, this study is unbiased since the authors question their previous conclusions and have recognized that the ‘main argument against a role for I_h channels has been their activation threshold which overlapped only slightly with the maximum diastolic potential’ (Ludwig et al., 2003) of SAN. Along with others, one of the earlier realistic experiments proved that the activation of funny HCN or I_Q current is not achievable within the pacemaking window of SAN, i.e., between the peak amplitude of oscillation that is often around -35 mV and MDP of -65 mV (Denyer and Brown, 1990). This is refreshing since one of these authors should be considered a pioneer in discovering the i_f (Brown et al., 1977).

Despite its strength, the aforementioned study exhibits some shortcomings. For example, in *figure 2D* (Ludwig et al., 2003), in contrast to the statement, the step did not hyperpolarize the neuron to -115 mV, and there is no sag with an amplitude of up to 2 mV, as most plausibly the MP trace is saturated at -100 mV for $\text{HCN2}^{-/-}$ (Fig. 17C). The latter occurs when inadequate high gain is used during the amplification. Those rebound activities are not ADP, especially in *figure 2F* (Ludwig et al., 2003), but RTP - rebound tail potential (Fig. 17A and B). Since the ADP is not generated in the absence of prior spikes. At least one spike is required for successive ADP, but it depends on the type of neuron (Fig. 17C). Not all neurons exhibit ADP, including the thalamocortical relay one, though this is erroneously stated (Fig. 17B). Thus, the latter is not an ADP but RAP - rebound action potential. For sure, the ADP does not occur both recurrently and sequentially three times, if the arrows were meant to represent it to potential readers (Fig. 17B). Data prove that the role of HCN2 is marginal for sag in these cells (Fig. 17A and B). Indeed, as stated, one of the prior studies addressed the slowly decaying ADP in relay neurons, but it occurred after 20 spikes evoked at 4 Hz (Lüthi and McCormick, 1998).

Nonetheless, the wild-type ($\text{HCN2}^{+/+}$) thalamocortical neurons exhibit a clear hallmark of HCN currents already at hyperpolarization by the -90 mV step (Fig. 18A), while the $\text{HCN2}^{-/-}$ only in response to the -130 mV step from a HP of -40 mV in *figure 2A* (Ludwig et al., 2003). The latter is consistent with the half-maximal activation voltage ($V_{0.5}$) of -88 vs. -115 mV (■ and □, Fig. 18C), respectively. Even if there was a correlation between the absence and presence of sag (see Fig. 17B), it is unclear why the neurons of both groups exhibit rebound action potential (RAP). It is not plausible that HCN2 is responsible for sag, while HCN1 triggers the RAP. Although based on assumptive expectation, HCN2 is absent in $\text{HCN2}^{-/-}$ mice, not comparing it with that of WT in *figure 5* (Ludwig et al., 2003) is puzzling along with the legend title of 'expression of HCN channel transcripts in wild-type and HCN2 knockout brain' (Ludwig et al., 2003). Instead, HCN1 , HCN3 , and HCN4 are compared. Since HCN2 should be absent while the expression of HCN1 and HCN3 is low, the recorded currents might occur via HCN4 . Dogmatically, the expressions of all these HCN α subunits at the mRNA level remain unchanged even during chronic ethosuximide treatment, and that of HCN2 is again not compared. It is established that the downregulation of one subunit triggers a compensatory mechanism via others. Also, during and over 4 weeks - aging, the expression of certain types of HCN could change.

Only at rest a dysrhythmia - an increased standard deviation of RR intervals (SDRR) - was estimated for $\text{HCN2}^{-/-}$ mice, yet the HR was comparable with that of WT (Ludwig et al., 2003). This is puzzling, since the I_h is present in diverse neurons of the CNS and in the SAN of $\text{HCN2}^{-/-}$ mice, albeit at lower amplitudes. Also, in the absence of current density estimation, any conclusion is without merit. Moreover, the range of MP between -140 and -90 mV for HCN current activation during patch-clamp recording in SAN cardiomyocytes is not that of the pacemaker, especially when the HP was -40 mV as in *figure 6D* (Ludwig et al., 2003).

Normalized values are not from activation (Ludwig et al., 2003), but from deactivation of channels - tail currents at constant -80 mV in thalamocortical relay neurons in a slice derived from mice (X, Fig. 18A and C). The 'activation curve' or voltage-dependent

activation for knockout HCN2^{-/-} mice is incorrect, as normalization in *figure 2C* (Ludwig et al., 2003) occurred in respect to WT for relay (□ and ■) and CA1 pyramidal neurons (○ and ●). Thus, the differences based on $V_{0.5}$ cannot be determined from the graph (Fig. 18C) for HCN2^{+/+} and HCN2^{-/-} mice. The I_h as presented in *figure 2A* (Ludwig et al., 2003) cannot be ‘nearly absent’ at 20.9 % (84 vs. 401 pA) of the control group at -130 mV. However, again, this comparison requires an estimate of the corresponding I_f current densities. Note that the latter is then accurate since it ‘applies only to functional channels, which are readily activated during experimental paradigms’ (Kodirov, 2023a).

In regard to *figure 3A* (Ludwig et al., 2003), the identical data set is unexpected and therefore without merit. Especially if the RMP of the HCN2^{+/+} relay neuron in the presence of 1 mM Cs⁺ was -83 ± 1 vs. -83 ± 1 mV after 100 μ M ZD7288, since data are not shown. Nearly the same applies as to HCN2^{-/-} groups, with respective values of -84 ± 1 vs. -84 ± 2 mV. The assumption that ‘tonic (or single spike) activity in which action potentials occur singly at lower frequency relatively independently of one another’ (Ludwig et al., 2003) is inadequate. Since a step evokes a train of APs, they follow one another and are thus interrelated. The term tonic is not equal to a single spike. In recorded traces, there is no sign of spontaneous spikes, whether in tonic or bursting modes. Why should Ca²⁺ transients underlie spontaneous spikes? How is it possible that 1 mM Cs⁺ enables the oscillatory behavior of transients while the same compound blocks the spindle periodicity (Lüthi and McCormick, 1998) via an identical target, HCN?

The aforementioned study (Ludwig et al., 2003) relies on a prior study on relay neurons (Lüthi and McCormick, 1998). The latter, however, uses non-physiological conditions, i.e., a strong hyperpolarization for up to 15 s. Not that the neurons are incapable of generating rebound spikes intrinsically, at least not repeatedly and with certain periodicity. Thus, applying such a paradigm is already meaningless since only spiking occurs recurrently at a certain frequency depending on actual behavior (e.g., feeding) but not hyperpolarization. The latter may occur for a short period without periodicity, at least not at a frequency that is observed during a train of spikes. Besides, the representative traces in *figure 2A* (Lüthi and McCormick, 1998) are inadequate, as the rebound activity occurs independent of the magnitude of injected currents, and they are two-fold apart at -200 vs. -400 pA. The design of the protocol and description of the results are incorrect, as the membrane potential response in the expanded time scale reveals that the timing to peak hyperpolarization was ~ 3 s, and there is not a single rebound Ca²⁺ spike but a burst. The same protocol was also used to modulate the properties of HCN currents in the thalamocortical cells of adult ferrets in *figure 4* (Lüthi and McCormick, 1998). Interestingly, these neurons have unusually bigger tail currents (Fig. 19A and B). Priming with a -2 nA $\cdot 5$ s step in current-clamp mode does not influence the subsequently recorded HCN currents in thalamocortical sagittal slices under voltage-clamp conditions (Fig. 19A). However, traces for the -104 mV step are missing. Also, the current amplitude and duration, are excessive even for adult neurons.

Clearly, one cannot apply 2 nA to neurons in whole-cell patch-clamp recordings for seconds (Kodirov et al., 2014). Although it is possible during conventional electrophysiology with sharp micropipettes, the statement that ‘electrode resistances ranged between 80–120 M Ω for current-clamp recording and 50–90 M Ω for voltage-clamp recordings’ (Lüthi and

McCormick, 1998) is confusing since both should be performed in one setting and with a single neuron. Next, instead of priming with a single step, the HCN channels were activated immediately after $-2 \text{ nA} \cdot 20 \text{ pulses at } 4 \text{ Hz}$ (Fig. 19B). For this purpose, the membrane potential was kept at -49 mV in current-clamp mode. However, the assumption that this MP value could prevent the activation of I_h is incorrect, since the extent of hyperpolarization by 2 nA pulses is excessive and either a single or cumulative sag is triggered in *figure 4A, B* (Lüthi and McCormick, 1998). After 120 ms of stimulation, the amplitude of I_h is bigger than the 'control' (Fig. 19B), which simply means that a recovery occurred during up to 15 s . Thus, again, the design of the experiments is inadequate. The brain most probably does not experience such periods of recurrent hyperpolarization. Besides, no one knows if the total durations, i.e., with 120 ms pulses vs. the 5 s step, are adequate for proper comparisons. The effects of differential priming are substantiated by values derived from tails after test potentials (Fig. 19C). Now, values are included for missing traces at the -104 mV step. Besides, at -59 mV , there are no HCN or tail currents; therefore, including a data point of absolute $\pm 0 \text{ pA}$ is incorrect. Interestingly, there is an additional outsider at either -49 or -50 mV . Therefore, the Boltzmann fit is slightly compromised. The same concerns apply to data points during repetitive stimulations (Fig. 19D). Also, it is unclear why, at -100 mV , the ratio should increase. It is possible that because of faster kinetics the tails are contaminated by capacitance currents. At least it is difficult to clearly distinguish the end of capacitive transients from the beginning of tail currents (Fig. 19A and B).

If arrhythmia indeed occurs subsequent to a seizure, a recent device could substantiate it (Inoue et al., 2017). The device allows cooling of the epidural surface of the cortex, and at 15°C , the EA was reduced in both cats and monkeys. Effects on EA were selective and reversible to a greater extent. Interestingly, the cooling did not affect the baseline amplitude of spikes that remained stable at $\sim 20 \mu \text{V}$ among two species and four groups.

A knock-down of VAPA decreases the HR in zebrafish to ~ 100 vs. $120 \text{ beats} \cdot \text{min}^{-1}$ of VAPB (Silbernagel et al., 2018). The combination of VAPA and VAPB slowed the HR to 60 vs. $150 \text{ beats} \cdot \text{min}^{-1}$ in control zebrafish at 72 hpf (hours postfertilization). Note that the VAPA was more efficient than the VAPB in HR modulation and vice versa when the I_h was compared. In double morphants, the effects on HR were heterogeneous, and even an atrio-ventricular conductance block was documented. The 2:1 and 3:1 atrio-ventricular block is also observed in $\text{Kcne5}^{-/-}$ mice (David et al., 2019). The HR in $\text{VAPB}^{-/-}$ is slower at ~ 560 vs. $620 \text{ beats} \cdot \text{min}^{-1}$ in *figure 5G* (David et al., 2019), and the effect is accompanied by T segment elevation. However, the corresponding values are ~ 670 vs. $730 \text{ beats} \cdot \text{min}^{-1}$ in *figure 5I* (David et al., 2019). The QTc in $\text{VAPB}^{-/-}$ mice has increased to ~ 49 vs. 45 ms , and therefore 'in the future, detailed regional action potential studies are indicated in $\text{VAPB}^{-/-}$ mice to further investigate this phenomenon' (Silbernagel et al., 2018), but it will be challenging to correlate such a small difference with respect to APD.

It is established that any agent, including anti-epileptic drugs (AED), may have unwanted side effects (Kodirov, 2023b). Moreover, it is possible that non-epileptic tissue may be sensitive to some AED, e.g., carbamazepine, comprising its side effect, while epileptic brain is resistant (Evans and Dougherty, 2018). The EA resembles, whether recorded in acute or organotypic slices, CA1 or CA3 pyramidal cells, and is induced either by 12

mM extracellular K^+ or deprivation of serum in culture medium (Ghasemi et al., 2019). Therefore, it led to the hypothesis that ‘each EA is also self-regulated by neurons and lasts for certain durations’ (Kodirov, 2023b).

Carbamazepine, which is also a Nav inhibitor, at 100 μ M significantly decreases the APs rate of pyramidal neurons in dorsal compared to ventral CA1 (Evans and Dougherty, 2018). Although many parameters of single spikes were comparable, the representative neuron in the dorsal hippocampus exhibited robust and fast AHP. It is anti-dogma that none of these pyramidal neurons exhibit spike frequency adaptation (SFA) even during 10 s steps, which is perhaps related to relatively large current injections of up to 330 pA. A clear adaptation occurred in DHC neurons only in the presence of carbamazepine and was independent of whether 250 or 330 pA currents were applied. Although the enhancement of ‘spike frequency adaptation across the longitudinal hippocampal axis’ by carbamazepine was concluded for DHC, it was only in 2 vs. 1 neuron in the ventral hippocampus. Statistics for $n = 2$ and comparisons are not adequate. Besides, the baseline train of spikes was evoked by 150 pA, while in the presence of carbamazepine, it was 240 pA. Although the DHC neurons were more sensitive to carbamazepine, the classical effects - Nav1 inhibition and reduced AP amplitudes - are also taking place in VHC (though the Δ of 3.5 and 3.8 mV was not significant). The α and β subunits of Na^+ channels, Nav1.1, Nav1.2, Nav1.3, Nav1.6, and β 1–3 were expressed in both DHC and VHC, while β 4 was in the former only.

The transfection with VAPB increased the contraction of HL-1 cells from 158 per min^{-1} to 179 (Silbernagel et al., 2018), yet did not increase I_h amplitude, though it to some extent altered activation properties. The voltage-dependency of activation is completely different in *figure 6G* (Silbernagel et al., 2018) compared to both oocytes and neurons, as even at -20 mV, the I_h is still available by $\sim 50\%$. Therefore, the $V_{0.5}$ of about -90 vs. -80 mV is inconsistent. The contraction rates of HL-1 cells were adequately measured again by providing another control group for a new set of experiments. Subsequently, a decrease in contractions from ~ 140 to 113 per min^{-1} was revealed by shVAPB (Fig. 9A). The effects of shVAPB are expected, but now even $\sim 60\%$ I_h is still present at -20 mV (Fig. 9B). Note that a holding potential was set to -30 mV in the HL-1 cell. Therefore, a pulse to -20 mV is not a hyperpolarization, and subsequently, the amplitudes are not derived from the activation of funny currents, and their polarities should be positive (Fig. 9B). Interestingly, all data points and the two fitted curves are overlapping, but still, statistical power was reached (Fig. 9C). If it is correct, then shVAPB is not only reversing the effects of VAPB but independently influencing the I_h properties. Speculatively, it could also reflect that there is always a tonic inhibition of I_h by VAPB, and a knock-down by shVAPB reveals an activation occurring earlier at -96 mV. Without doubt, it is a challenge to conduct electrophysiology in beating cells, a condition that could contribute to the magnitude of the obtained I_h values. Note that only 30% of HL-1 cells exhibit I_h currents, with the threshold for activation occurring at -60 mV and the $V_{0.5}$ at -83 mV (Sartiani et al., 2002). In this case, the HP of -40 mV was comparable to that of the abovementioned study.

Of note, the ‘HL-1 cell line was isolated from a culture of AT-1 cells after more than 100 separate preparations and attempts to passage AT-1 cells *in vitro*’ (Claycomb et al., 1998). As the name AT-1 suggests, it is a mouse atrial cardiomyocyte tumor lineage.

Therefore, the statement that the HL-1 cells are a ‘spontaneously beating sino-atrial node-like cardiomyocyte cell line’ (Silbernagel et al., 2018) is incorrect. Note that the founders did not present the observed APs and do not even mention the word ‘node’ (Claycomb et al., 1998).

Also, robust and unique outward currents in HL-1 were characterized as I_{Kr} -like (Claycomb et al., 1998). These currents exhibit the HERG phenotype, especially the bigger tails at -50 mV than the delayed rectifier component during depolarization from a HP of -50 mV in native cardiomyocytes. Both components are obliterated by $10 \mu\text{M}$ dofetilide, a class III antiarrhythmic medication. Recordings after 100 % confluency (Silbernagel et al., 2018) reflect more realistic conditions, as ideally, the HL-1 cells may form a tissue monolayer and communicate as pseudosyncytium. However, it is then difficult to control the voltage and space clamp.

9. HCN and behavior

HCN channels were first described in a relatively simpler organ, the *cor* - heart, with clear-cut regions and designated cells. Later on, in CNS, the HCN-mediated currents were also registered. Interestingly, the phenotypes of currents in cardiac and brain cells resemble, especially those of rabbit sino-atrial node and LS neurons (Fig. 1B and C).

However, the brain is complex, and e.g., a targeted recording of nucleus *ventralis basalis* (VB) neurons is not easy in vitro, even if they were fluorescent. The thalamus, in contrast to the cortex or hippocampus, does not have laminae, and the VB alone possesses the ventrolateral (VL), ventral posterolateral, and posteromedial (VPL and VPM) subdivisions (Paxinos and Franklin, 2001). This brain atlas proves that distinguishing borders with neighboring nuclei is possible only schematically (Fig. 12).

Besides, if the I_f is robust (Solomon et al., 1993), there is no need to use excessive pharmacology in order to isolate with concurrent $3 \mu\text{M}$ TTX, 10 mM TEA, 0.5 mM BaCl_2 , 0.1 mM CdCl_2 , $50 \mu\text{M}$ AP-5, $20 \mu\text{M}$ CNQX, and $50 \mu\text{M}$ bicuculline. Note that at least two of these compounds in turn influence the I_h (Araque et al., 1995; van Welie et al., 2005). Although adequate comparisons are not possible, the waveforms of currents and potentials are different from prior study in VB (Rateau and Ropert, 2006; Rivero-Echeto et al., 2021). If HCN channels are present, then the I_h and sag in the respective membrane current and potential often correlate (Fig. 6). The sag in MP also depends on the intensity of hyperpolarization from RMP (Fig. 6A). Interestingly, in this LS neuron during voltage-clamp experiments, the HCN was activated only upon -140 mV steps (Fig. 6B, C, and F). An opposite was observed in VB neurons, i.e., in the absence of sag, robust HCN activation occurred already at -90 mV (Fig. 8A and B). Perhaps in Fig. 8A, the -120 pA step was not sufficient to activate the required numbers of HCN α subunits. This notion, however, is disproved by the saturation of MP amplitudes. Nonetheless, the presence of rebound action potential hints that HCN channels are activated. In Fig. 8B, the TTX at a standard $1 \mu\text{M}$ concentration was used; therefore, spontaneous AP-independent neurotransmission ($\mathcal{J}$) still occurs superimposed, which however does not interfere with waveforms of HCN.

The amplitude and waveform of HCN are also sensitive to addictive drugs and thus could theoretically modulate underlying behavior. However, a concurrent administration of caffeine and cocaine into mice will not reflect the PACO - coca paste effects in men, even if the assumption is supported by many select and cross-referenced studies (Rivero-Echeto et al., 2021). Why not feed the mice with PACO? Why do mice need a warm environment after an *i.p.* injection of caffeine and cocaine? Was the body temperature checked after cocaine? As the latter may cause acute hyperthermia, rats will require a colder environment. Home cages are usually kept at a standard and constant temperature, along with other parameters.

The number of unprotected head-dipping (toward the room floor from the locations of the open arms) and rearing - upright posture on two hind limbs is different (Pacheco et al., 2021). The mean values appear apart at $\sim 4.205 \pm 0.535$ vs. 2.448 ± 0.438 in the crack-cocaine group (Fig. 15C, arrow reflects the pairs of non-overlapping SEM). There is some degree of significance based on $\sim 4.20 \pm 0.53$ vs. 5.47 ± 0.61 for the respective male and female controls (Fig. 15C). The same applies to the frequency of rearing between males and females (Fig. 15D).

An increase in open field rearing numbers was not considered a significant increase in spontaneous locomotion (Pacheco et al., 2021). Although the data was thought to be comparable between groups, the legend to *figure 5* (Pacheco et al., 2021) states that 'PN-crack females had a shorter latency time for immobility' (Pacheco et al., 2021). Why should immobility, in the course of the forced swim test, during the last 4 min be significantly longer in naïve females vs. males? In other words, why do control females have depressions? The same applies to long-term memory consolidation estimated by the step-down avoidance test. *Figure 9A* (Pacheco et al., 2021) is completely incorrect for females, as the laminae of the *cornu ammonis (CA1)* area of the hippocampus are not visible, and the background without neurons is not different from that of the amygdala. Why should the *alveus* become fluorojade-positive (FJ+) and represent neurodegeneration? These results prove the opposite (Castro et al., 2011), as 0.6 mg·µl pilocarpine injected into the hippocampus does damage other nuclei. Since crack cocaine 'exposure in females alters *striatal* cells and *serotonergic* function, which likely mediates the behavioral deficits' (Pacheco et al., 2021). The latter is also biased, as the 'male rats did not present increased susceptibility to seizures, and since FJ+ is associated with seizures, these animals were not analyzed'. However, pilocarpine causes seizures in both sexes, which should result in neurodegeneration. Finally, it is better not to use 'these women are often marginalized' or 'considered hypersexual deviants' or 'crack cocaine babies' (Pacheco et al., 2021). It is not very common that a woman starts to use crack cocaine only after being pregnant and then exhibits the abovementioned hyperactivity during pregnancy. Even if marginalizing behavior occurs, then why in the first place, based on cocaine use or else?

The changes after ZD7288 are excessive in *figure 2C* (Rivero-Echeto et al., 2021), though the 30 µM concentration is not high. The reason is perhaps the higher IR of this neuron, which could not cope with injected currents. This is substantiated by the unstable behavior of the MP response close to the end of the 500 ms · 100 pA pulse. The latter step is not a 'hyperpolarizing input' (Rivero-Echeto et al., 2021), and in the brain, the term *input* has different meanings. The increase in IR by ZD7288 is also dramatic, and therefore,

the amplitude, duration, and phenotype of low-threshold spikes (LTS) are altered. Thus, the rebound spike cannot be considered an LTS. The ‘action potentials triggered by a LTS were called low-threshold action potentials (LT AP) and those triggered [sic] later were called high-threshold [sic] action potentials (HT AP)’. But the same sentence with some modification appeared in legend of *figure 7* (Rivero-Echeto et al., 2021): ‘action potentials triggered [sic] by a low-threshold [sic] spike (LTS) were called low-threshold action potentials (LT AP) and those triggered [sic] later were called high-threshold [sic] action potentials (HT AP)’. Finally, ‘note how in the presence of Ni^{2+} , an unmask [sic] component related to I_H current deactivation also contributes to rebound bursting’. However, in *figure 2B* (Rivero-Echeto et al., 2021), there is no bursting, and only a single LTS is depicted, which is prevented by Ni^{2+} without affecting the sag or related rebound tail potential (RTP).

The association of cocaine with local anaesthetics (von Anrep, 1880) is not unfounded, as it reduces I_h in a similar manner to bupivacaine, lidocaine, and mepivacaine (Bischoff et al., 2003). The I_h remained unaltered when QX-314 was applied externally to small neurons within the DRG slices, thus proving that this quaternary derivative of lidocaine is hydrophilic. Although the R(+) enantiomer exerted similar effects on I_h with an inhibitory concentration (IC_{50}) of $55 \pm 6 \mu\text{M}$, it is hard to obtain an identical mean value of $55 \pm 5 \mu\text{M}$ for bupivacaine.

Note that, up to date, only ZD7288 is reliably considered a selective antagonist of HCN. However, the instantaneous currents at -100 mV from a holding potential of -40 mV in *figure 1A* (Rivero-Echeto et al., 2021) are not affected by $30 \mu\text{M}$ ZD7288, which is unexpected. The behavior of currents upon return to HP of -40 mV is unusual, and those are not tails, as ZD7288 does not alter them, but perhaps an AC - action current (Fig. 13E). The AC, in this case, is an artifact that underlies action potential. This hallmark is also observed in control recordings with redundant protocol - heterogeneous pulse durations - in *figure 1B* (Rivero-Echeto et al., 2021), as the range between -40 and -70 mV is not hyperpolarization. That is, there are two components: faster capacitance and slower AC. Although the tail of HCN is always an inward current under physiological conditions, with this protocol, it should exhibit two polarities and decay much faster (Araque and Buño, 1994). Therefore, for comparisons with prior and closely related studies, not 500 ms but longer tail pulses are adequate (Rateau and Ropert, 2006).

In naïve controls, the SE occurs at $0.6 \text{ mg}\cdot\mu\text{l}$ pilocarpine (Pacheco et al., 2021), and susceptibility increases at $1.2 \text{ mg}\cdot\mu\text{l}$ (Fig. 16A and B). A lower dose reduces the latency-increased susceptibility in crack cocaine-exposed females but not males (Fig. 16A). At $1.2 \text{ mg}\cdot\mu\text{l}$, effects were considered significant in males but not females (Fig. 16B). The latter dose reduces susceptibility in females (Fig. 16D, $\sim 1648 \pm 139$ vs. $2021 \pm 215 \text{ s}$). Why should all values be lower at $1.2 \text{ mg}\cdot\mu\text{l}$, even for controls of the same sex? For female controls, the values at $0.6 \text{ mg}\cdot\mu\text{l}$ apparently were $\sim 4784 \pm 421$ vs. $1648 \pm 139 \text{ s}$ at $1.2 \text{ mg}\cdot\mu\text{l}$, which constitutes a 65 % decrease (Fig. 16C and D, black). Therefore, because of the dramatic (not logarithmic) dose-dependent effects of pilocarpine and an unknown amount of cocaine exposure (especially since pups are not from the same mother), no firm conclusion could be drawn based on these results.

Cocaine targets monoamine uptake transporters, Cav, HCN, and Nav channels. Therefore, 'maternal crack cocaine use in rats leads to depressive- and anxiety-like behavior, memory impairment, and increased seizure susceptibility in the offspring' (Pacheco et al., 2021). Subsequently, many studies addressing the inhibition of DA uptake also use amfonelic acid, methylphenidate, and nomifensine, comparing their effects to those of cocaine alone. Cocaine, similar to amfonelic acid, methylphenidate, and nomifensine, triggers stereotyped gnawing (Tirelli et al., 1998), which could be considered a sign of hyperactivity. Nomifensine prevents both experimental axonal degeneration and DA level lowering in nigrostriatal neurons (Kolacheva and Ugrumov, 2021). Rats more frequently press a lever for food pellets (reinforcement) when receiving amfonelic acid, cocaine, methylphenidate, and nomifensine (Poncelet et al., 1983).

Even if the cocaine effects on HCN were plausible, it still has many other targets. During electrophysiological experiments, it is possible to isolate the involvement of HCN, which should occur during voltages reflecting terminal repolarization or hyperpolarization. The HCN role during depolarization is minimal. Besides, the upper limit of the ramp at 800 pA is excessive (Rivero-Echeto et al., 2021), therefore spikes are transitioning into spikelets. Note that the response of only acute group includes a slow afterhyperpolarizing potential (AHP) upon termination of ramp. What underlies this difference? There is no known link between AHP and HCN. A fast AHP is present after each spike and its amplitude is gradually lower upon progressive depolarization. None of group exhibit low threshold spikes under these conditions and those considered as LTS, are reflecting a simple amplitude adaptation occurring during double-spiking or burst-like activities. Only the lower amplitude and longer duration of spike may prove that there was LTS, but it is not quantified.

10. HCN and pacemaking dogma

Earlier studies postulated the role of distinct additional currents in spontaneously excitable cells besides those selective for K^+ or Na^+ . Although at the same time more skeptical studies were conducted, nowadays the pacemaker time pattern of AP generation is attributed solely to HCN.

HCN is expressed in several tissues, but the bulk of research is conducted in the heart. Note that evidence of the presence of HCN was available prior to one electrode patch-clamp. As 'after having impaled three microelectrodes simultaneously, constant current pulses were applied through one electrode near the end of the strip and the electrotonic potentials at two different sites were recorded' (Noma and Irisawa, 1976). These recordings are qualitatively and quantitatively comparable with most modern traces. However, since the presented original one current and two potential traces were overlapping, for clarity, they were redrawn and separated (Fig. 20). In this tissue sample, the MDP reached about -67 mV, while the overshoot was -10 mV. In SAN tissue, the two underlying currents, inward HCN and outward K, were also registered (Fig. 21). Importantly, based on these particular recordings, a hint for the presence of HCN conductance was provided. As the 'greater the magnitude of hyperpolarization, the larger and faster was the change in magnitude of the inward current' that is in contrast to outward K conductance. The τ of the outward current deactivation - K tail was on average 370 ms upon return to HP of -40 mV. Authors have tested this HP, since

back then it was established that the RMP of the SAN was lower than the MDP, i.e., -38 vs. -67 mV. It was modestly concluded that the inward 'conductance changes may not be solely attributable to the deactivation and activation of the potassium current as in the Purkinje fiber' (Noma and Irisawa, 1976), since the amplitude was gradually higher during the longer hyperpolarization and inward current tails were faster than the outward one. Therefore, the presence of multiple types of ionic currents during the hyperpolarization was suggested. In SAN, the reversal of current polarity, i.e., the 'rectifier property shown in the Purkinje fiber' (Noble and Tsien, 1968), might also be involved in slow depolarization after a quiescent period' (Noma and Irisawa, 1976).

It has been documented that the isolated SAN cells of New Zealand white albino rabbits exhibit spontaneous pacemaker activity with an MDP of -65 mV (Fig. 22A). The threshold for AP was around -35 mV (Denyer and Brown, 1990); therefore, this value was chosen as a HP, and the MP was changed to -65 mV for 500 ms by 10 mV incremental steps (Fig. 22B). The latter protocol was able to activate only ~ 50 pA inward currents upon the step to -65 mV (Fig. 22C). Note that it is still a mild 30 mV change in MP, and one could perhaps use either ± 0 mV or even the peak overshoot range of $+30$ mV as a HP, assuming that the HCN activation is initiated already during fast MP decline right before the actual MDP. In any of these scenarios, changing the MP to -65 mV is not a hyperpolarization but rather a repolarization, as this is an MDP and corresponds to RMP. This particular point even illegitimizes the name of the channel - hyperpolarization activated. Although this elegant experiment revealed a clear cut I_f in the same SAN cardiomyocyte upon hyperpolarization to -95 mV (Fig. 22C). Note that similar Cs^+ -sensitive currents were distinguished from I_M and termed I_Q in neurons (Halliwell and Adams, 1982), though it was not explained which word underlies the letter Q. In *figure 3* (Halliwell and Adams, 1982), the muscarine at $20 \mu\text{M}$ concentration obliterated the 'relaxation' of inward currents activated from a HP of -40 mV since it activates the muscarinic acetylcholine receptor (mAChR). Although the 'relaxation' of inward currents, when activated from a HP of -70 mV, resembled, they were not sensitive to muscarine under identical conditions.

Based upon experiments, the permeability of the Q-channel to K^+ and Na^+ was mathematically modeled (Edman et al., 1987). The same slow inward currents were termed the I_S component, but they are not solitary channels contributing to the pacemaker function of SAN (Yanagihara and Irisawa, 1980).

11. Conclusions

Since this millennium, the extent of interest in HCN channels has increased significantly, starting shortly after several publications in the late XXth century (Santoro and Tibbs, 1999). Main progress has been achieved by the patch-clamp method (Hamill et al., 1981; Kodirov, 2020, 2023b). The functional presence of now-established HCN - sag in membrane potential in true pacemaker tissue was elegantly shown earlier. As the 'man-made S-A node strips 0.3 mm in width and longer than 0.5 mm showed spontaneous action potentials of normal shape and amplitude' (Noma and Irisawa, 1976).

Since the birth of the HCN channel, the term for underlying currents has evolved greatly and a plain *funny* is not the original one (Brown et al., 1977; DiFrancesco and Ohba, 1978; Noble and Tsien, 1968; Santoro and Baram, 2003). Collectively, ‘as one may gather from the terms *funny* and *queer*, the question as to the precise role of I_h in cellular activity has left many investigators scratching their heads’ (Luthi and McCormick, 1998).

Nonetheless, not all cells exhibit I_f and even not the most appropriate pacemakers, SAN cardiomyocytes, at least not in the expected physiological range of MP (Denyer and Brown, 1990). A precise experiment substantiated that HCN channels do not significantly activate in the physiological MP range of pacemaker cells. In SAN cells, a threshold for I_f activation upon pulses to -65 mV was achieved, though the latter is a maximal operating range of negative MP - MDP. Nonetheless, the elucidation of HCN currents using the patch-clamp method is still insightful and may help to characterize the effects of medications.

It is perhaps adequate to discern the role of HCN mutations within the scope of clinical studies, but ‘causative epilepsy-linked mutations appear to be rare in patients’ (DiFrancesco et al., 2019). Even if both the idea and impact of heteromerization are driven by earlier studies on seizure ontogeny, the resulting opposite regulations of HCN1 and HCN2 or HCN1/HCN2 heteromere (Brewster et al., 2005) and the rationale for expected effects could be distinct from just functional alteration in a dominant-negative manner. The same is true as ‘despite the potential of iPSC-CMs, immature, embryonic-like phenotype and large intrapopulation heterogeneity limit their ability to model adult cardiomyocyte physiology’ (Schroer et al., 2019) and cure arrhythmias or compensate for HCN mutations. Clearly, the morphology of action potentials in LQTS2 iPSC-CM with a transient upstroke is not ventricular-like (Pahlavan et al., 2018).

Also, data are still scarce with regard to the ancillary proteins of HCN and their possible effects on I_h activation. Even if ‘VAPB modulation of HCN channels might be highly relevant for our understanding of many diseases such as cardiac arrhythmias, epilepsies, inflammatory or neuropathic pain, and ALS’ - a familial form of amyotrophic lateral sclerosis (Silbernagel et al., 2018), based on ‘results and data from the literature, it is difficult to draw a clear genotype-phenotype correlation between the dysfunction of the HCN channels and epilepsy’ (DiFrancesco et al., 2019). Finally, since a single inhibitor of HCN, ivabradine, is used for angina, anticonvulsant, antioxidant, arrhythmia, and neuroprotective therapies, targets are clearly not solely funny currents (Cavalcante et al., 2019).

There is no absolute selective or devoid of side effects antagonists of HCN channels. The same is true for HERG K channels, but at least many class III antiarrhythmic drugs are established and used frequently. Despite the use of ivabradine in clinical trials and the fact that it blocks the I_f in the SAN of guinea pigs (BoSmith et al., 1993), there is still an ongoing search for potential antagonists. Therefore, perhaps another three decades will be required to assure the efficacy of carboxamides on HCN1, though a facilitated summation of EPSPs and perhaps correlated modulation of working memory in animals were observed (Harde et al., 2024). The EPSP summation and learning plasticity occur by participation

of designated neurotransmitters. It remains unanswered why the HCN1 should remedy the related abnormalities in humans, even potentially.

The HCN downregulation perhaps indirectly may promote epilepsy via alteration of neuronal excitability, but one of the unresolved questions is how this leads to LQT when there is no role for I_f in the repolarization phase of AP. The gap in our knowledge is that, if one wishes, there is an expected answer for anything, as ‘an altered T-wave morphology observed in the ECGs of VAPB^{-/-} mice supports the recently proposed role of HCN channels for ventricular repolarization’ (Silbernagel et al., 2018). Since the role of HCN is crucial in pacemaker cells and regions, while the repolarization is possible only by and via Kv channels, and there is no unique association, axis, binding, link, pathway, or target, the future directions will be to perform experiments adequately and publicize them precisely. In this case, the universality of phenomena will become a solid fact at least. Only then will experiments on flies, mice, rats, and snails have translational applications.

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Data availability

All visualized items are available from the corresponding author upon request.

Abbreviations:

ADP	after-depolarizing potential
AED	anti-epileptic drugs
AHP	after-hyperpolarizing potential
AP	action potential
cAMP	cyclic adenosine 3',5'-monophosphate
cGMP	cyclic guanosine 3',5'-monophosphate
DHC	dorsal hippocampus
EA	epileptiform activity

HCN	hyperpolarization-activated cyclic nucleotide gated non-selective cation channel
HP	holding potential
I_h	currents via HCN (or pacemaker channel, since it was discovered in SAN)
iPSC-CM	induced pluripotent stem cell-derived cardiomyocyte (hiPSC-CM, from human)
KA	kainite
LA	lateral amygdala
LS	lateral septum
MDP	maximum diastolic potential
MP	membrane potential
PACO	coca paste, leaf extract of <i>Erythroxylum coca</i> containing psychoactive alkaloids and cocaine (hence <i>paco</i> - pasta de cocaína in Spanish)
PP	pacemaker potential
PP	place preference
RAP	rebound action potential
RMP	resting membrane potential
RTP	rebound tail potential
SAN	sinoatrial node
SFA	spike frequency adaptation
SWB	spikes within burst
$V_{0.5}$	half-maximal activation or inactivation voltage, same as $V_{1/2}$
VAMP	vesicle associated membrane protein
VAPA	VAMP associated protein A
VAPB	VAMP associated protein B
VAPC	VAMP associated protein C
VB	nucleus <i>ventralis basalis</i>
VHC	ventral hippocampus
VL	ventrolateral VB

VPL	ventral posterolateral VB
VPM	ventral posteromedial VB

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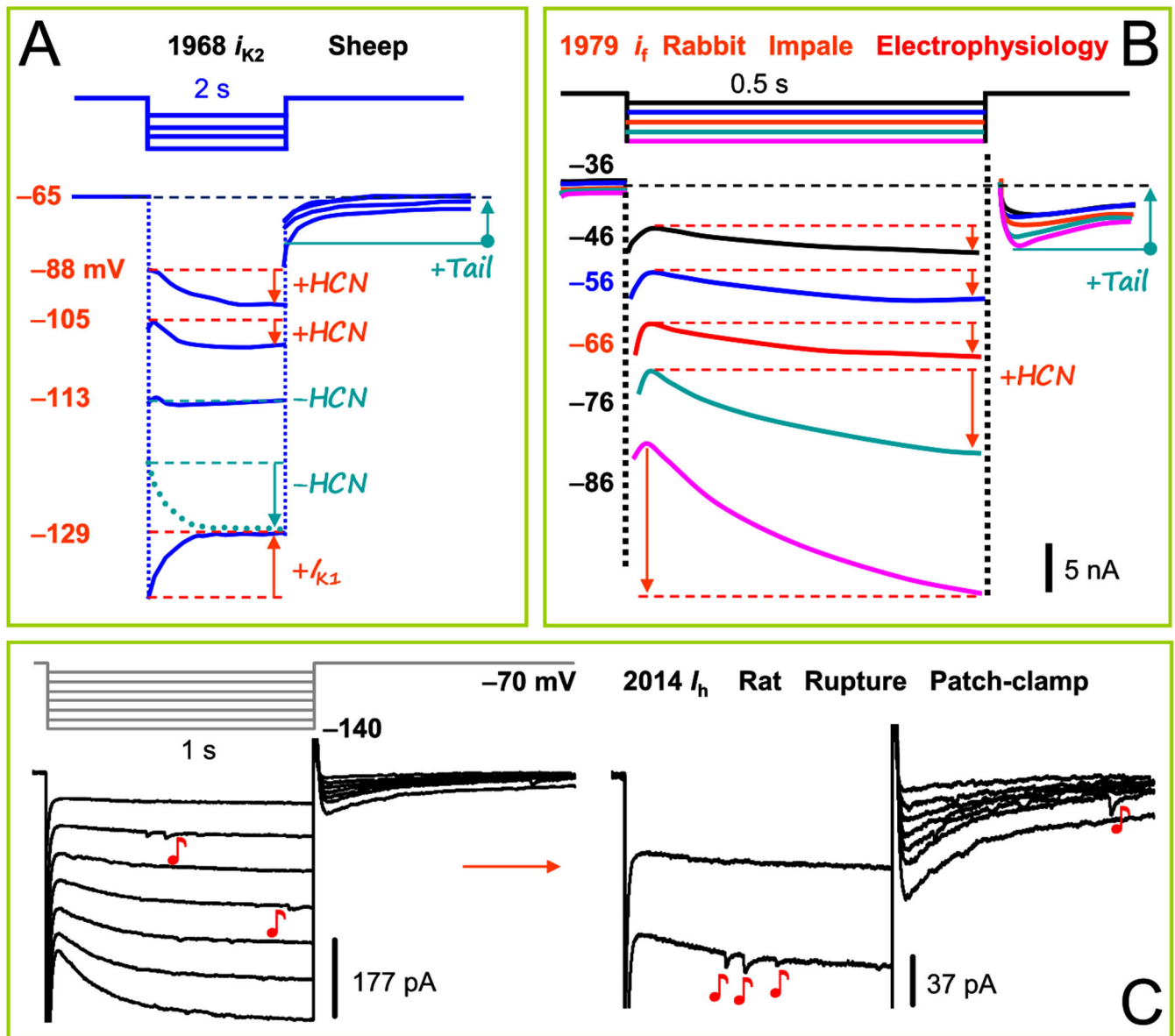


Fig. 1.

Very first hints for HCN channel presence. (A) Gradually activating inward currents termed i_{K2} and resembling the I_f (+HCN) in sheep Purkinje fiber. Modified *figure 5* (Noble and Tsien, 1968). Recordings were obtained at a relatively low 2 mM concentration of extracellular KCl. The duration of steps was 2 s, and HP was -65 mV, but the amplitude scale is missing. The level of hyperpolarization is indicated next to the corresponding traces. At -88 and -105 mV, typical I_f (+HCN) was activated, which was absent at -113 mV. At -129 mV, the activated currents started immediately to inactivate with distinctively faster kinetics and are presumably of K_{ir} origin (+ I_{K1}). The I_{K1} is perhaps masking the I_f currents since, upon the termination of each step, comparable deactivation (+Tail of HCN) of hyperpolarization-activated currents occurred in a voltage-dependent manner. Most plausibly, the absence of I_f (-HCN, dotted theoretical trace) at -129 mV could be only

visual, and the underlying funny currents may be unmasked by blocking the I_{K1} component. (B) Very first recordings of typical currents in rabbit SAN termed I_f but not yet derived from the established I_f for *funny*. Modified *figure 16* (Brown et al., 1979). It was simply termed in accord to an ‘additional current’ in frog sinus venosus tissue (see Fig. 2). The HP was set to -36 mV that closely corresponds to maximum diastolic potential, which was perhaps crucial to record a typical current phenotype (+HCN) in contrast to mixed ones in A. (C) The same typical phenotype is observed in neurons of the CNS, as exemplified by responses in the lateral septum (LS) slice. The latter is independent from modes of electrophysiology techniques, i.e., whether the cell is impaled with the sharp electrode or its membrane is ruptured with the suction. Since the RMP of the neuron was -63.9 mV, the activation of I_h was achieved at a holding potential of -70 mV. Note the presence of spontaneous EPSCs during hyperpolarized test potentials and I_h tails. Modified *figure 7* (Kodirov et al., 2014).

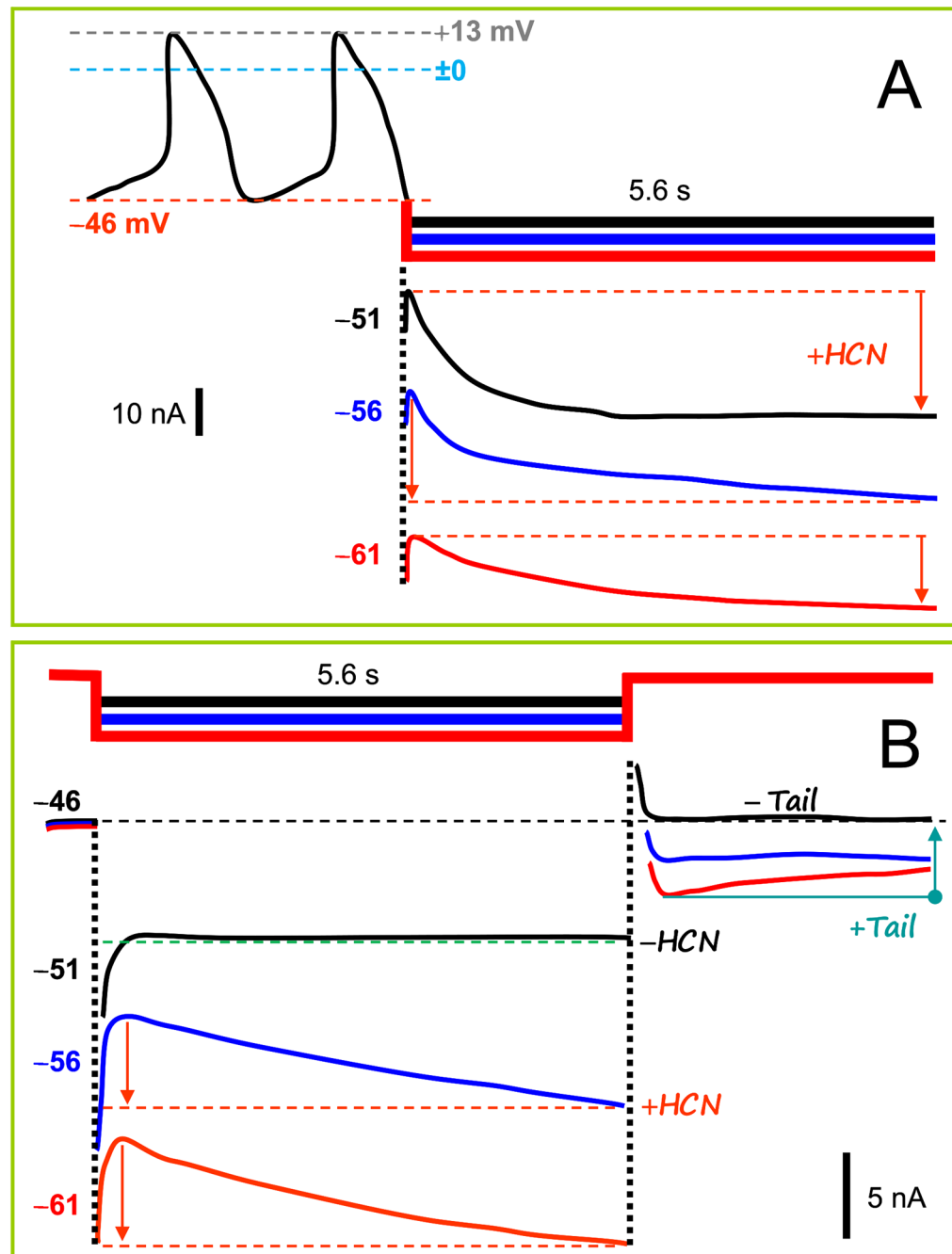


Fig. 2.

Pacemaker APs and HCN channel activity. (A) One of the first physiologically relevant recordings of I_f was termed simply as an additional current in frog sinus venosus tissue. After recording the spontaneous APs, the MDP was incrementally (5 mV) hyperpolarized each time for 5.6 s. The typical current phenotype ($+HCN$) is obvious at each step. (B) Similar experiments using the voltage-clamp protocol. Despite the similarity, the phenotype of currents is not identical to A. The HP was set to -46 mV, which is the value of the maximum diastolic potential in A.

Modified *figure 7*(Brown et al., 1977).

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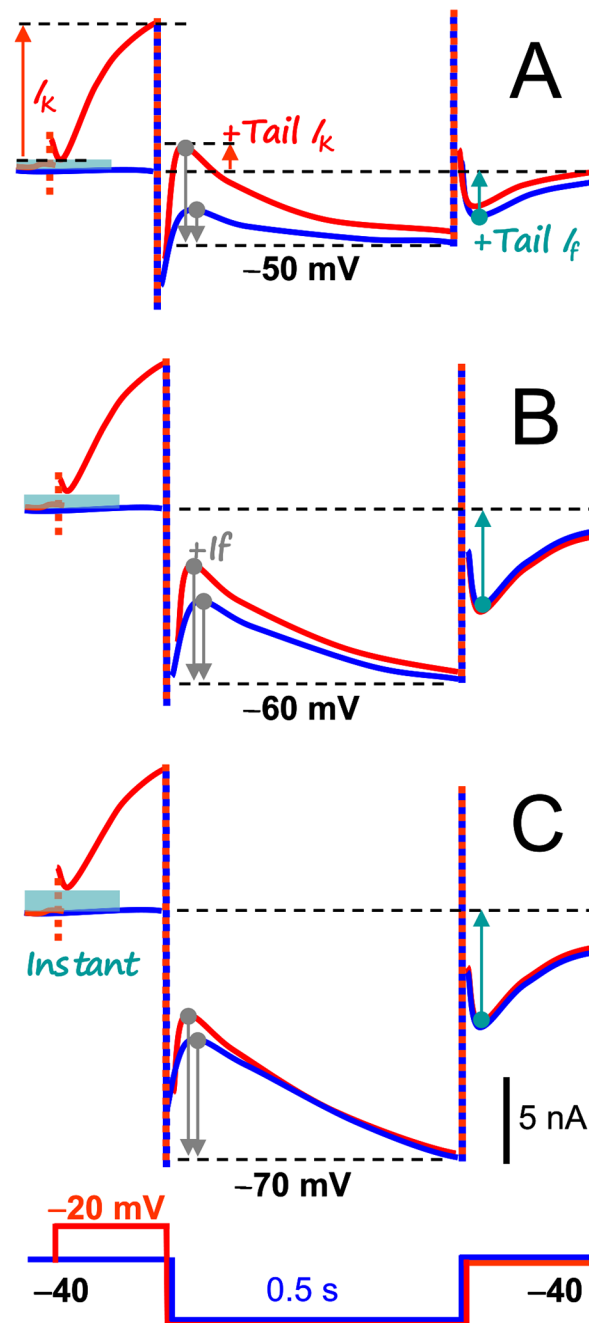


Fig. 3.

Membrane potential dependent influence of I_K on HCN channel activity. (A) Depolarization of MP by a prepulse to -20 mV from a HP of -40 mV activated I_K , and subsequent repolarization to -50 mV triggered i_f that gradually developed during 500 ms steps in rabbit SAN tissue (see protocol in C). Upon the return to HP of -40 mV, an inward tail of i_f was generated. In the absence of prepulse, the amplitude of i_f was smaller, and the entire current was inward. (B) Similar experiments, albeit with a test potential of -60 mV. (C) Subsequent test by hyperpolarization to -70 mV. Under these conditions, the prepulse to -20 mV had

no effect on i_f activation; the amplitudes at the end of the step are similar. Recordings are insightful and reliable, although a minor increase in instantaneous currents occurred upon the prepulse to -20 mV, which could be related to sequentially applied steps and increasing hyperpolarization. Also, note that the tails were less dependent on the prepulse of -20 mV than the actual HCN currents.

Modified *figure 4* (Brown and DiFrancesco, 1980).

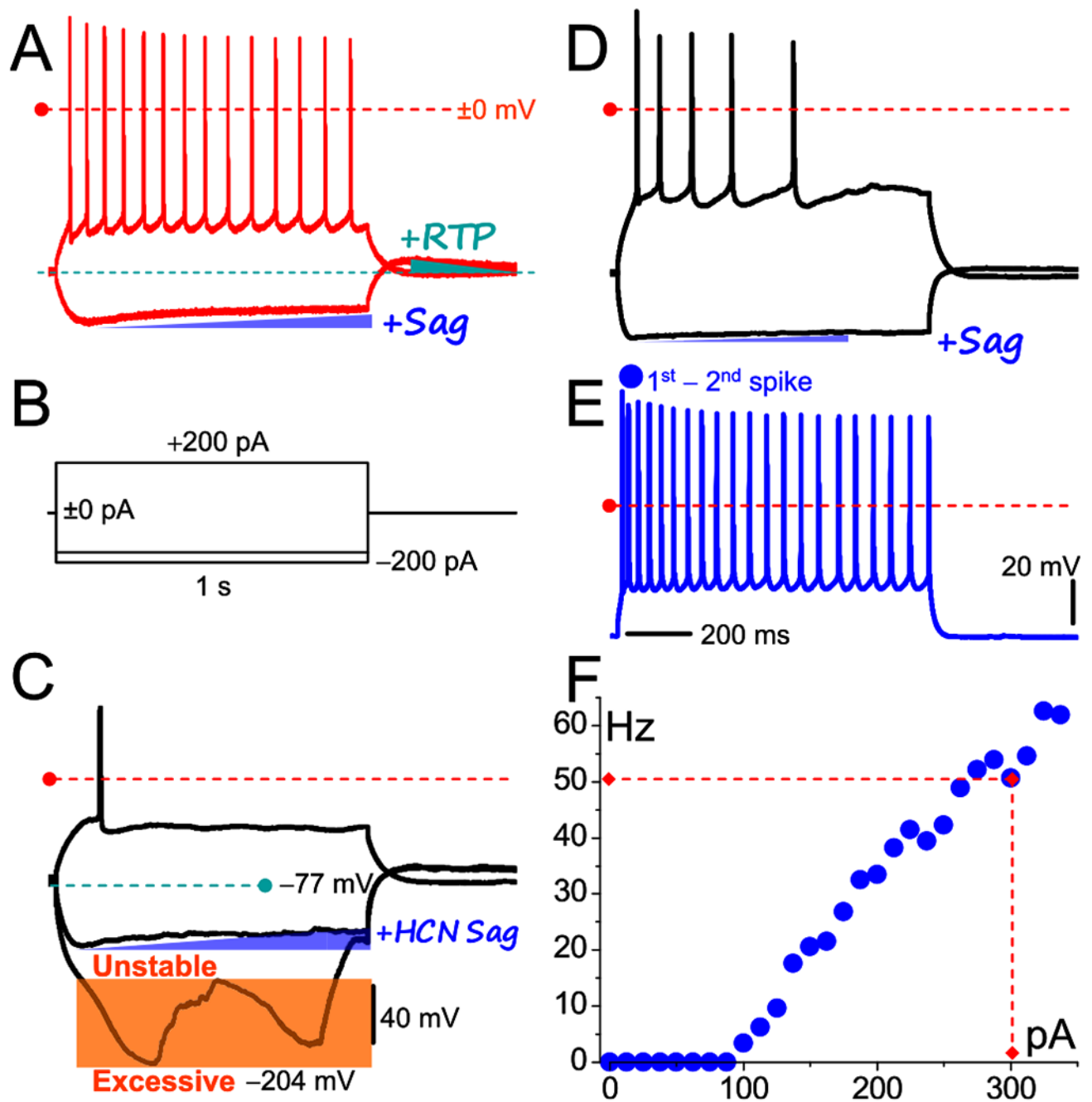


Fig. 4.

Sag in MP. (A) Response of adult cortical neuron to symmetrically injected ± 200 pA current. (B) Step protocol. (C) Temporary instability in MP by -200 pA in the lateral amygdala (LA) neuron. At the subsequent -160 pA step, the response was as expected, revealing a sag in MP. (D) Typical behaviour of the pyramidal cell of LA. (E and F) LA interneuron and instantaneous frequency at the start of the AP train for each step. The time intervals between the 1st and 2nd spikes were used for the latter estimation. The amplitude of sag was -6.1 , -9.8 , and -2.7 mV in the respective neurons in A, C, and D. In E, the

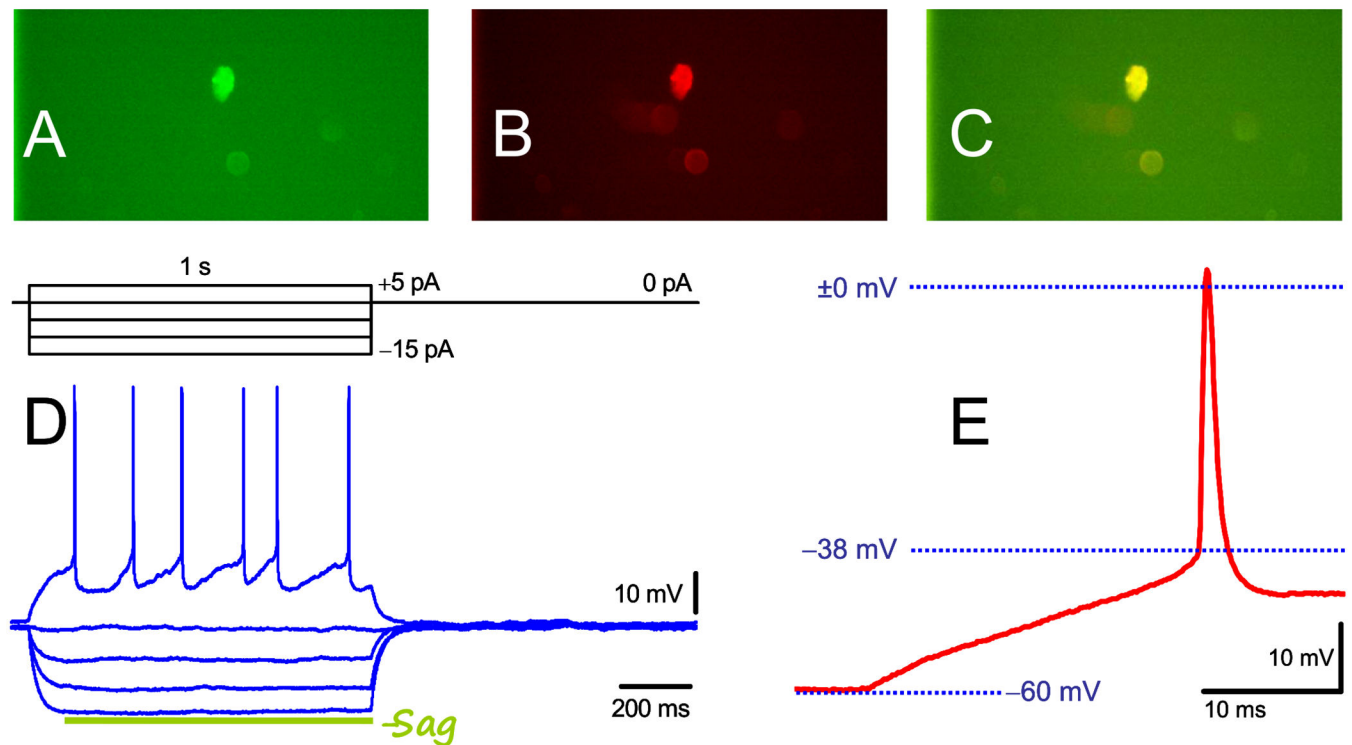
absence or presence of sag was not elucidated. Scale bars in E identically apply to A and D. RMP was stable at -61 , -77 , -59 , and -76 mV for A, C, D, and E (Kodirov, unpublished observations).

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**Fig. 5.**

Properties of immature neuron. (A) Targeted GFP⁺ neuron migrated to the LII/III of the cerebral cortex of the host mouse, which was around 3 weeks old. The host received the E17.5 embryonic GFP⁺ neurons by the injection into the right subventricular zone (SVZ) at P0. The GFP⁺ neuron was loaded with biocytin (B) during recordings as staining and merging reveal (C). (D) Marginal currents were able to significantly hyperpolarize the MP or trigger a train of spikes, as the R_{in} was ~ 1.6 G Ω . (E) The waveform and time course of the first AP along the RMP, threshold, and overshoot range with a peak at +2 mV at 31 °C are indicated in an expanded amplitude and time scale. The absence of neural terminals and high input resistance substantiate the immature phenotype of this cell (Kodirov, Kilb, and Luhmann, unpublished observations).

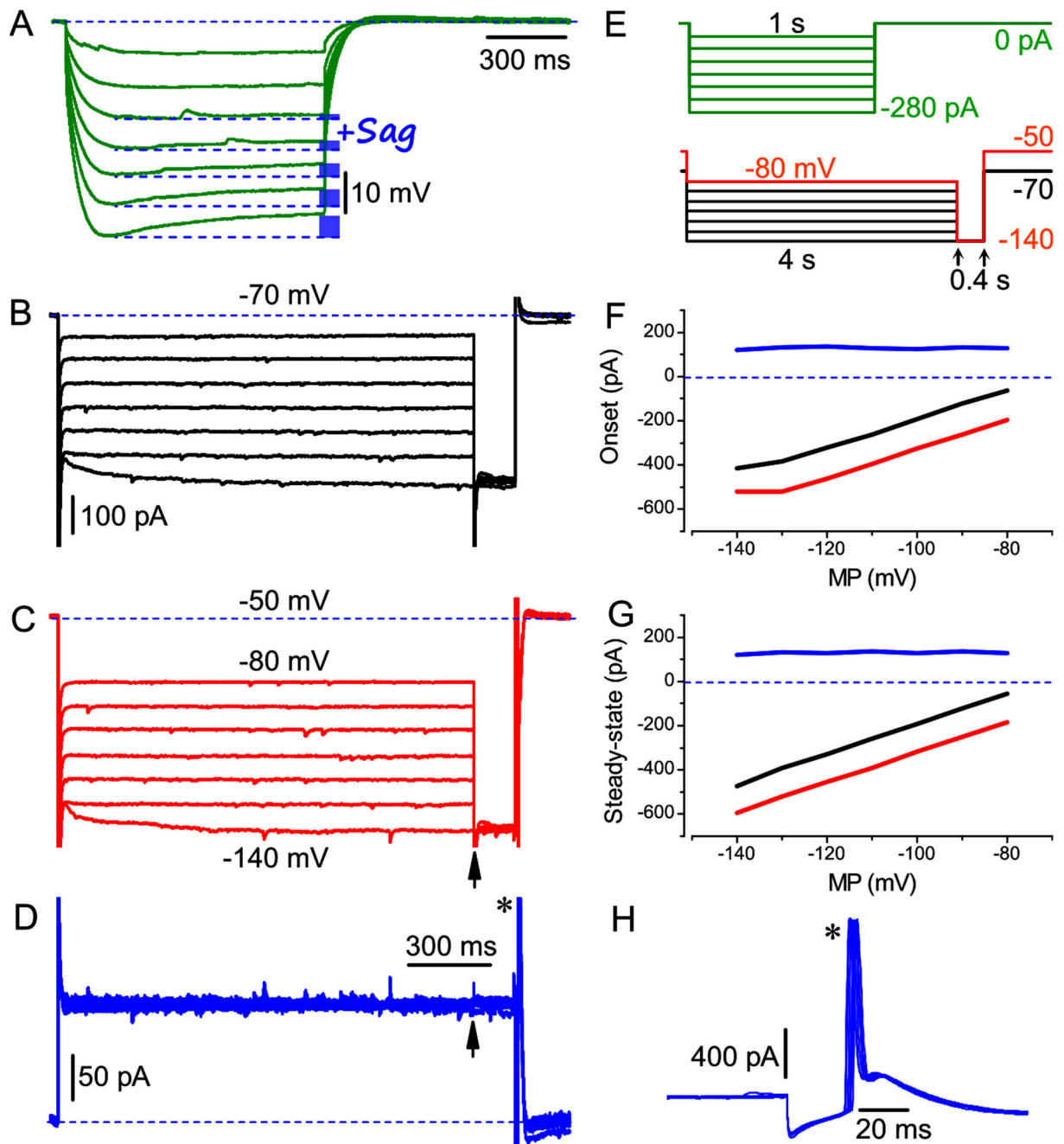


Fig. 6.

The holding potential independent activation of I_h in the lateral septum. (A) MP response to the hyperpolarizing current protocol shown in E. The value of the RMP of this neuron was -61.1 mV. (B) Subsequently recorded I_h at a holding potential of -70 mV in the same neuron. (C) The activation of HCN channels from HP of -50 mV. (D) The corresponding seven subtracted traces at tested potentials. (E) Current and voltage protocols with respective 40 pA and 10 mV increments. (F) The color-matched values of the instantaneous currents. (G) The steady-state amplitudes. (H) The portion of the subtracted currents in the expanded

time scale. Scale bars are shown for all traces. The time scale for B–D and the amplitude scale for B and C are identical. Note that the artifacts of RAPs appear inverted (Kodirov, unpublished).

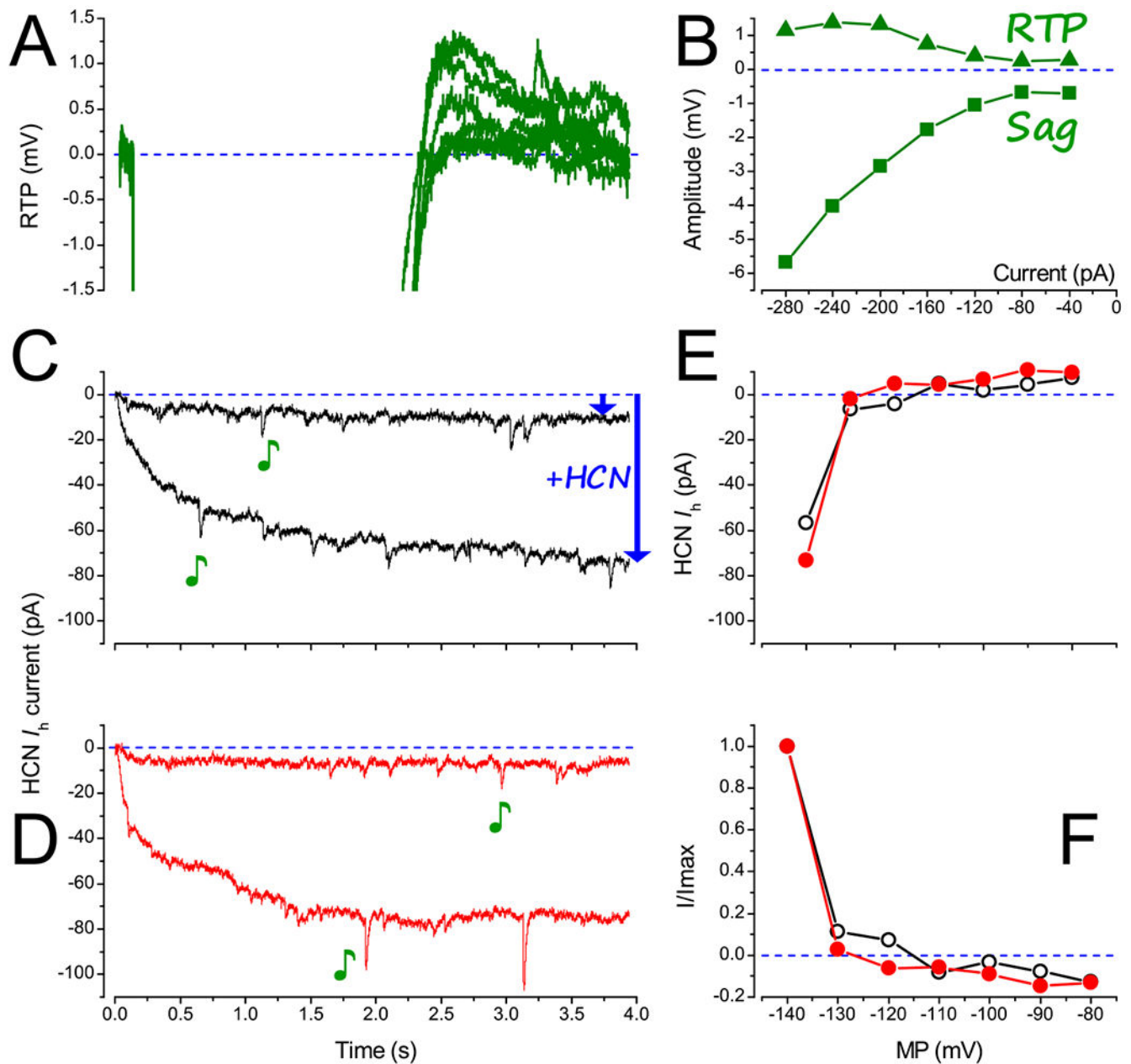


Fig. 7.

HCN-mediated inward current and sag in membrane potential. (A) RTP on an expanded amplitude scale. All traces are from Fig. 6. (B) The amplitude of RTP was small but correlated with the magnitude of sag in MP (estimated from A). (C and D) Respective HCN-mediated inward currents upon activation from a holding potential of -70 and -50 mV by steps of -140 and -130 mV. (E) Corresponding magnitude of the pure I_h , i.e., the difference between the amplitudes of currents upon the onset and those at the end of hyperpolarizing 4 s steps from C and D. Note that in this LS neuron only at -140 mV, robust and slowly equilibrating HCN-mediated currents were observed, which increased from 58.6 pA (O) to 73.2 pA (●). The values at all other tested potentials (-130 to -80 mV) were

roughly close to ± 0 pA. (F) Normalized currents reveal the presence of I_h at -140 mV and the inability of the subsequent -130 mV step to reactivate them significantly (Kodirov, unpublished).

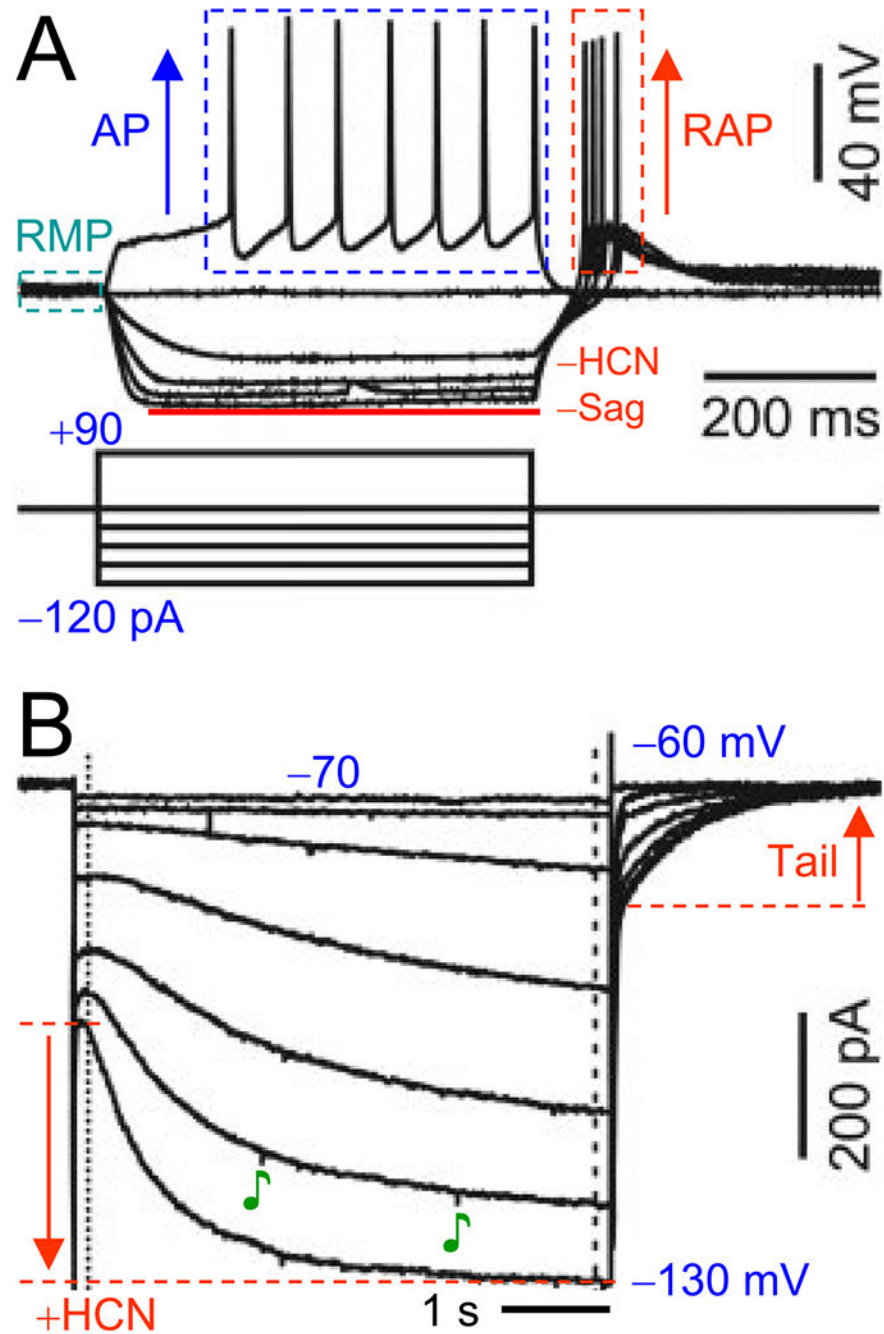


Fig. 8.

Neuronal membrane current and potential properties of the nucleus *ventralis basalis* (VB). (A) Response of VB neuron to 500 ms depolarization and hyperpolarization from RMP of -66 mV in whole-cell patch-clamp configuration. Since the slice is from the P11 mouse, the input resistance (R_{in}) is high at 745 M Ω . Note that the stated ‘anomalous rectification’ - sag in MP is negligible under these conditions. Upon depolarization, spikes are generated after a long latency. Termination of hyperpolarization leads to rebound action potential (RAP). Although the RAP is sometimes considered a slow Ca^{2+} or low threshold spikes (LTS), its

height is not apart from that generated by step depolarization. In this case, it is possible that the slow Ca^{2+} spike is obscured by the burst of fast Nav APs. (B) Same neuron in voltage-clamp mode. In contrast to A, robust HCN currents are progressively activated. The conditions are close to physiological, as the holding potential is set to -60 mV, which proximate the RMP of this neuron at -66 mV, and the maximal step to -130 mV is not excessive. Note that this simple protocol enables tails resulting from HCN openings during the prior 5 s and subsequent closure by termination of hyperpolarization. Based on response patterns, even 2 s would suffice to activate HCN. The $\mathcal{J}$ denotes the small PSCs which are not resolvable because of 5 s recordings but clearly is distinguishable during 500 ms as in A though only a single event occurred during hyperpolarizing steps. Vertical dotted and dashed lines are respective instantaneous and relative steady-state currents. Modified (Rateau and Ropert, 2006).

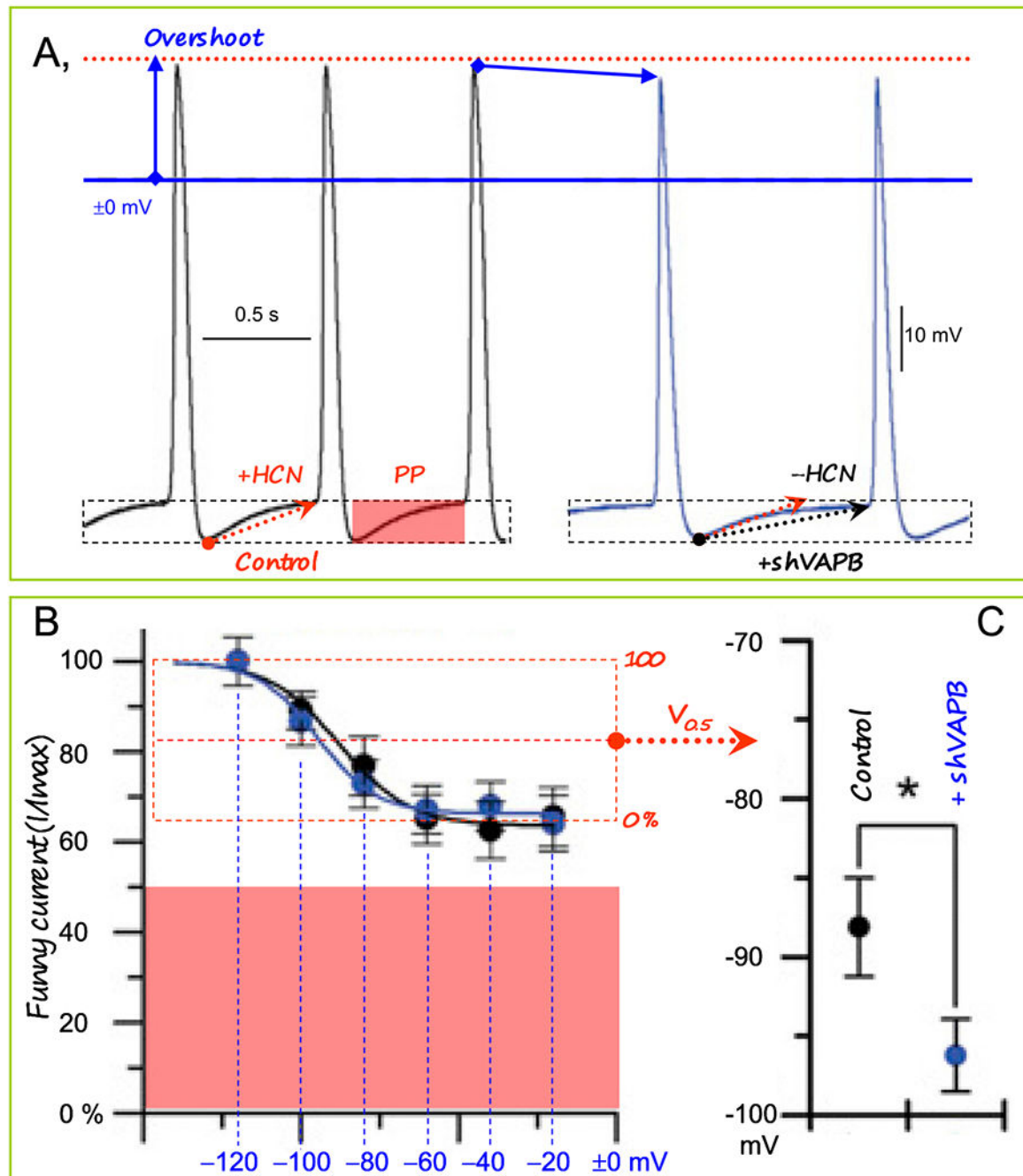


Fig. 9.

HCN and pacemaker potential (PP). (A) The decreased frequency of spontaneous APs in control and shVAPB-transfected HL-1 cells reveal the contribution of HCN. The presence of shVAPB did not alter the AP phenotype but slowed down the PP and thereby the frequency. This can be appreciated by the superimposed blue and red arrows. Note the small effects on peak overshoot potential that are perhaps because of additional effects on either Nav1 channels or since the PP reaches the plateau enabling stable RMP prior to AP. In control HL-1, the PP reaches the threshold for AP quickly, and there is only MDP but not RMP.

Whole-cell current-clamp experiments were conducted in a 5 to 125 mM K^+ gradient after 100 % confluency. (B) Normalized I_h in control and shVAPB-transfected HL-1 cells (black and blue symbols, respectively). Whole-cell currents were evoked from a HP of -30 mV. Although data points and SEM cross each other, a small and non-overlapping portion of the curves helped to conclude otherwise. (C) Resultant significant effects after fitting. Note that when recordings and analyses are adequate, the curves enabling the estimation of half-maximal activation voltage ($V_{0.5}$) would also include data points, ideally up to the 0 % current level (red background). Therefore, the $V_{0.5}$ was estimated artificially as shown in B with the arrow.

Modified (Silbernagel et al., 2018).

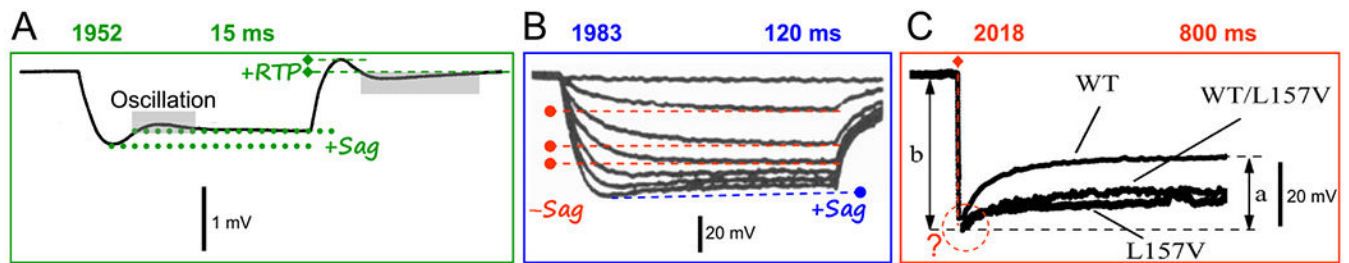


Fig. 10.

Evolution in the sag recording standard is dominated by the focus on genetics. (A) The oscillatory behavior of membrane potential upon $15 \text{ ms} \cdot 1.49 \text{ mA/cm}^2$ step in the giant axon of *Loligo* (Hodgkin and Huxley, 1952). Retrospectively, it is obvious that prior to the oscillations, the sag and related RTP were present despite extremely small hyperpolarization (see Fig. 4A). (B) The absence and presence of sag in olfactory cortex neuron (Constanti and Galvan, 1983). The absence of sag is independent of the shorter pulse duration of 120 ms (see also Fig. 6A). Logically, the sag was present at a stronger hyperpolarization achieved by current steps up to 1.5 nA. (C) Unusual recordings and presentation of representative responses to 800 ms $\cdot$ 350 pA steps in neurons of the neonatal cortex (Bonzanni et al., 2018). Since the traces are unnecessarily truncated, their durations might not be 800 ms. By estimation of the a/b ratio, the magnitude of 'sag' was proposed. Note that the traces should reveal the effects of mutation, as WT refers to the transfection of wild-type HCN1 channels. The other two combinations are L157V expression alone or concurrently with WT HCN1. Panels A, B, and C are modified from the respective aforementioned studies.

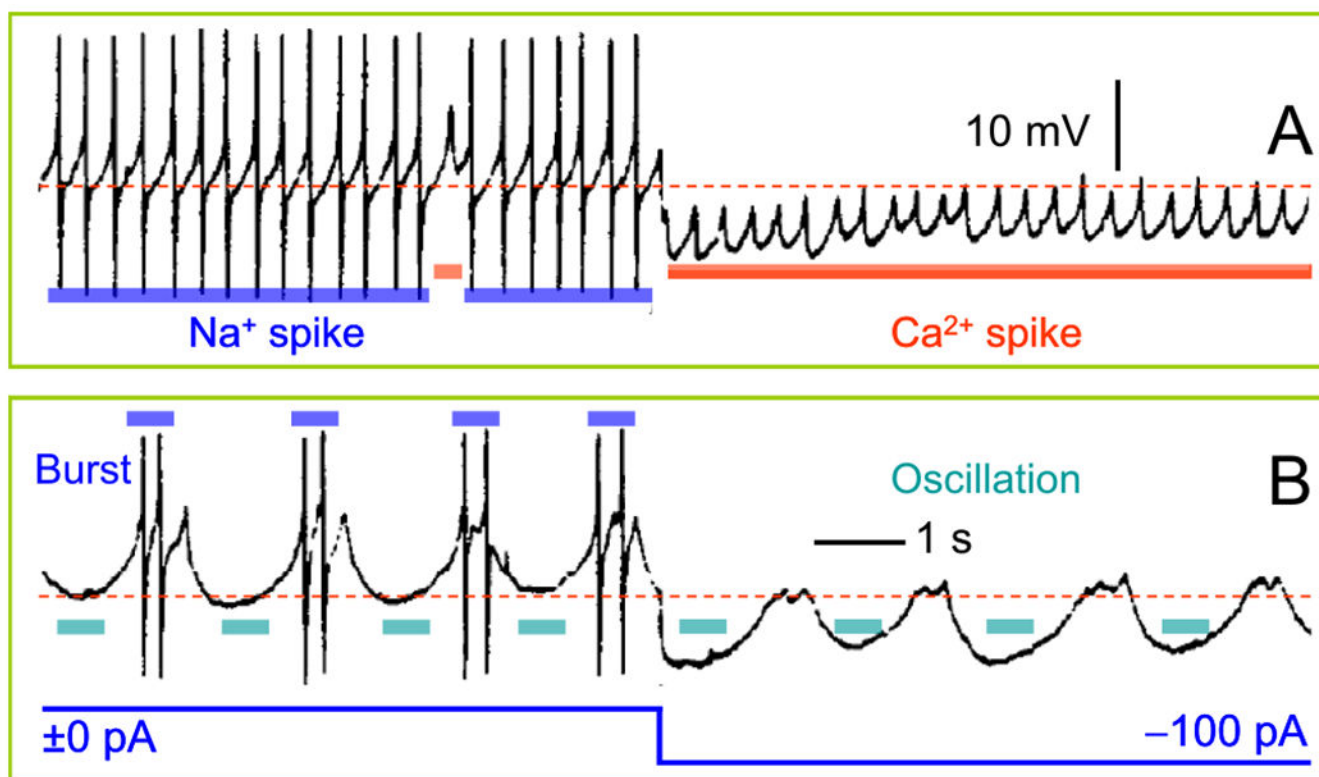


Fig. 11.

Cocaine effects on AP and MP. (A) Spontaneous Na⁺ spikes and a Ca²⁺ spikelet at RMP under control conditions in locus coeruleus neurons of P21 rat. Note a single Ca²⁺ spikelet at RMP while preventing Na⁺ AP upon step hyperpolarization by a -100 pA current injection. (B) Similar recordings in the presence of 10 μM cocaine clearly manifest several effects: Oscillations at a regular rate, bursting of Na⁺ APs during up-states, and obliteration of Ca²⁺ spikelets by a -100 pA step. The timing of oscillation allowed the precise generation of two spikes, though often the spontaneous MP depolarization has almost reached the threshold for 3rd AP but coincided with the start of downstate. These effects of cocaine are mediated via multiple ion channels and can be tested directly. During oscillations, the role of HCN channels may prevail.

Modified (Williams and Marshall, 1987).

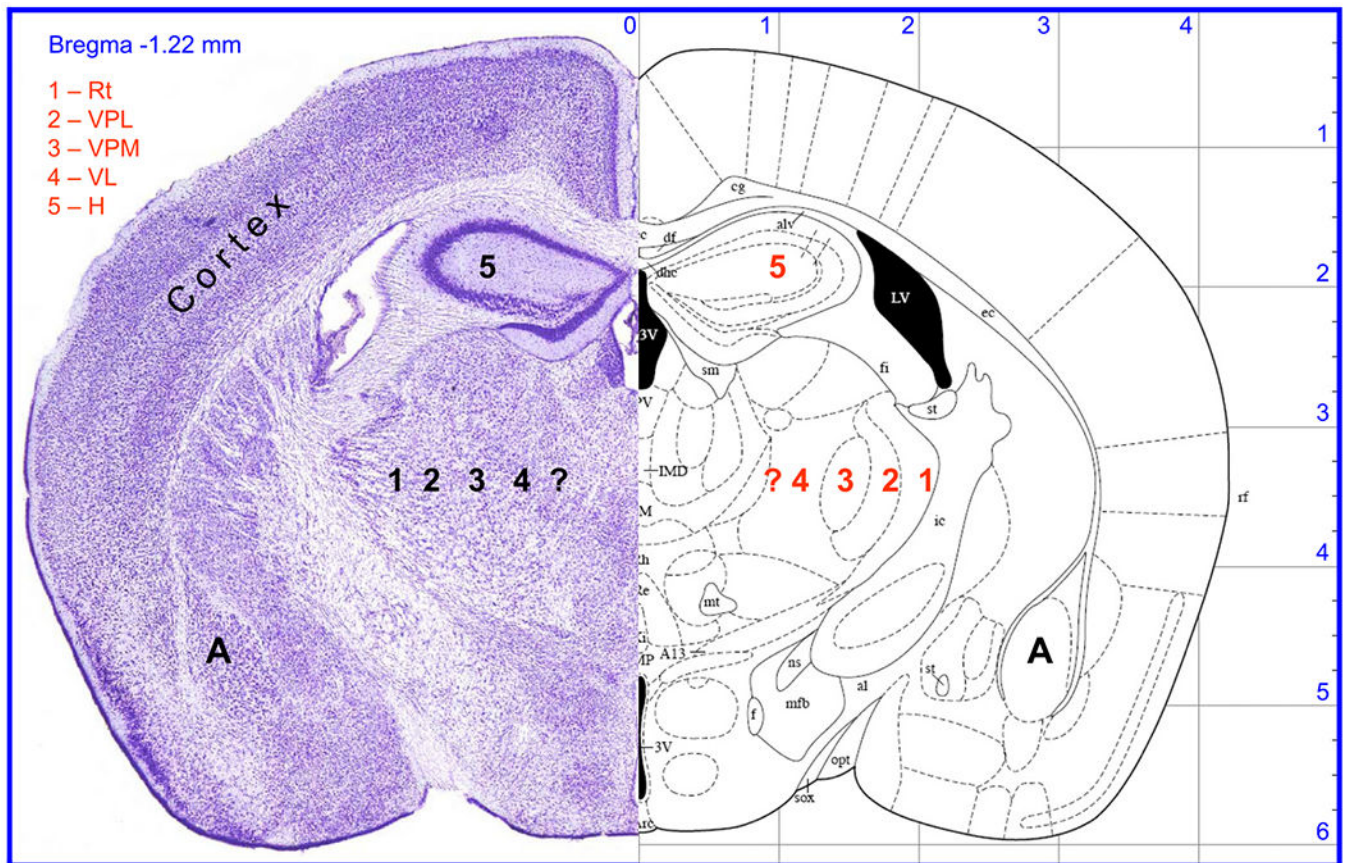


Fig. 12.

Location of the hippocampus and nucleus *ventralis basalis* (VB). Coronal microtome section of mouse brain and schematic distribution of nuclei and subdivisions. Even in rodents, discriminating between the borders of diverse nuclei of the thalamus is difficult. Especially since at different Bregma some of the subdivisions become smaller. How can one conduct a patch-clamp recording in the VB neuron in a targeted manner in each nucleus using a comparable appearing slice? Although a similar task in regard to the A - amygdala, cerebral cortex, and H - hippocampus is feasible to accomplish. Thalamic nuclei: Rt - reticular, VL - ventrolateral, VPL - ventral posterolateral, and VPM - ventral posteromedial. Modified (Paxinos and Franklin, 2001).

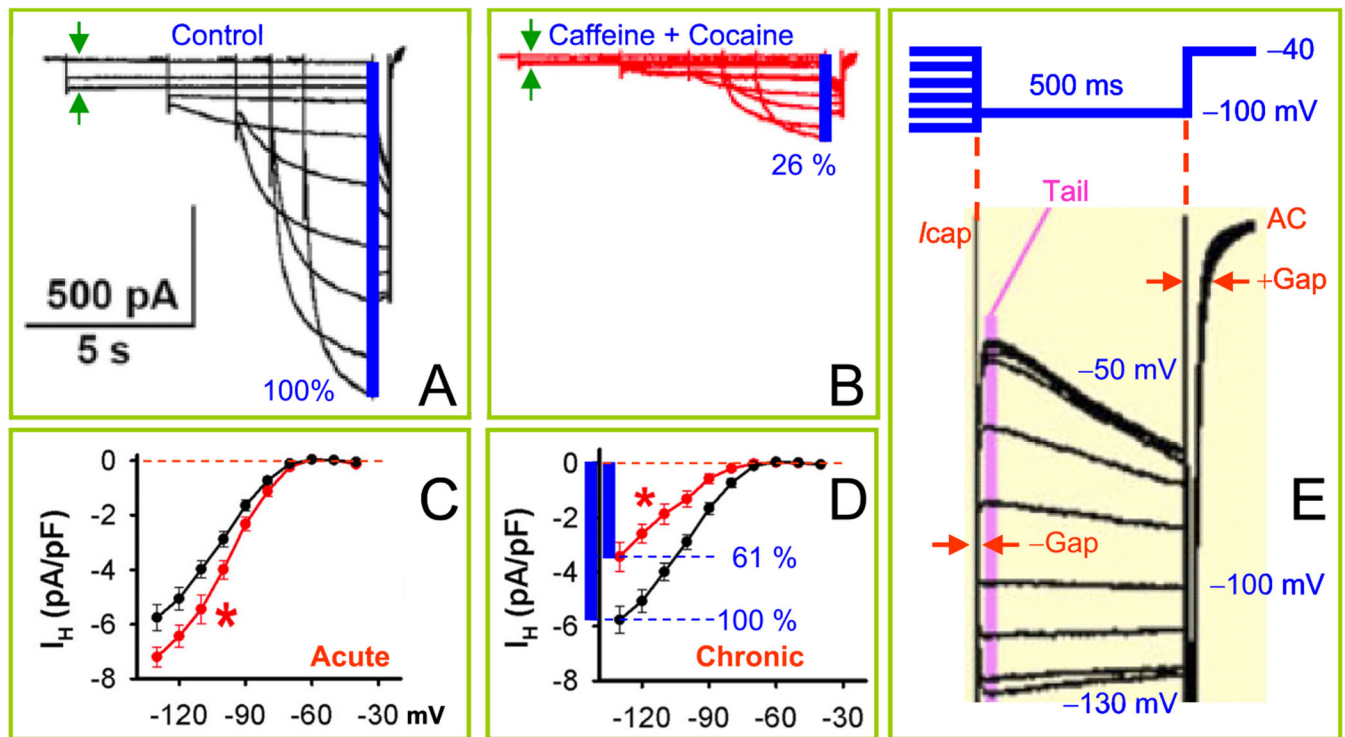


Fig. 13.

Concurrent caffeine and cocaine effects. (A) Membrane currents in a neuron of VB upon step hyperpolarization under control conditions. (B) Traces from a chronic caffeine + cocaine binge after 13 days. Obliteration of instantaneous currents at -50 and -60 mV test pulses is a side effect. Note ‘how PACO-like binge treated VB neurons presented less I_H ’ (Rivero-Echeto et al., 2021). (C) Mean peak values of the control group are in black ($n = 6$), while those of the acute 24-hour caffeine + cocaine binge are in red ($n = 5$). (D) The same control values are used for the chronic group ($n = 6$). Note the dramatic inhibition by 74 % in B that is much >39 % in D ($n = 5$). Since the n numbers are low, the latter difference should result in bigger SEM bars in contrast to the presented ones. Finally, if indeed the recording in B represents ‘average traces’ of $n = 5$, then the mean values in D are incorrect. (E) Membrane currents in a neuron of VB upon steps to identical -100 mV that is a designated tail pulse. Behaviors of tails hint that HCN were activated by prior hyperpolarization prevailing between -80 and -120 mV (see Fig. 15). At -130 mV, the amplitude of the tail starts to saturate. The wider gap between capacitance I_{cap} and membrane currents is perhaps derived from action currents (AC). Since termination of the -100 mV step leads to sufficient depolarization, It is an artifact of spikes but has an opposite polarity because of voltage-clamp mode.

Modified from respective *figures 1 and 6* (Rivero-Echeto et al., 2021) and highlighted.

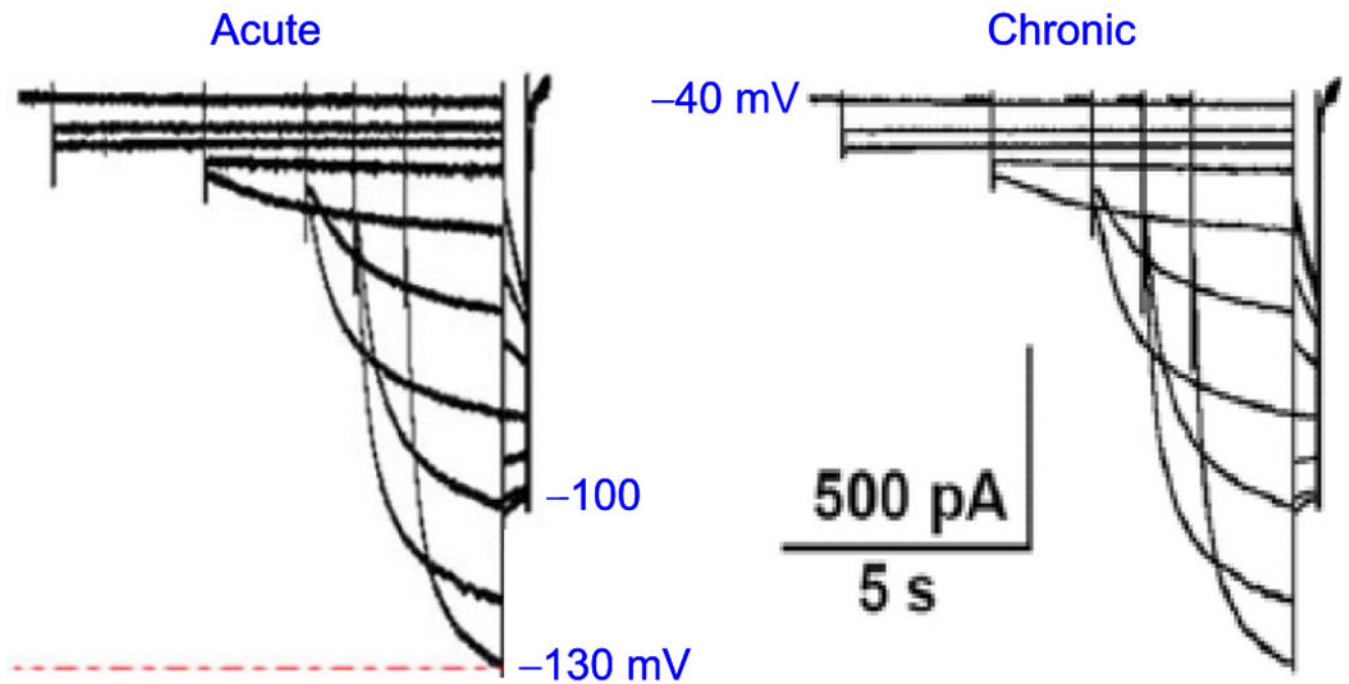


Fig. 14.

Identical traces for different conditions. Membrane currents in the VB neuron upon hyperpolarization. The same traces represent controls for the acute and chronic groups of caffeine + cocaine binge. Perhaps traces for the chronic group are low-pass filtered as capacitance transients are decreased. Since it is not clear why and how 'average current traces' were obtained for $n = 6$. A brain slice is from a mouse, sometimes after P25. Modified from respective *figures 1B and 6A* (Rivero-Echeto et al., 2021) and highlighted.

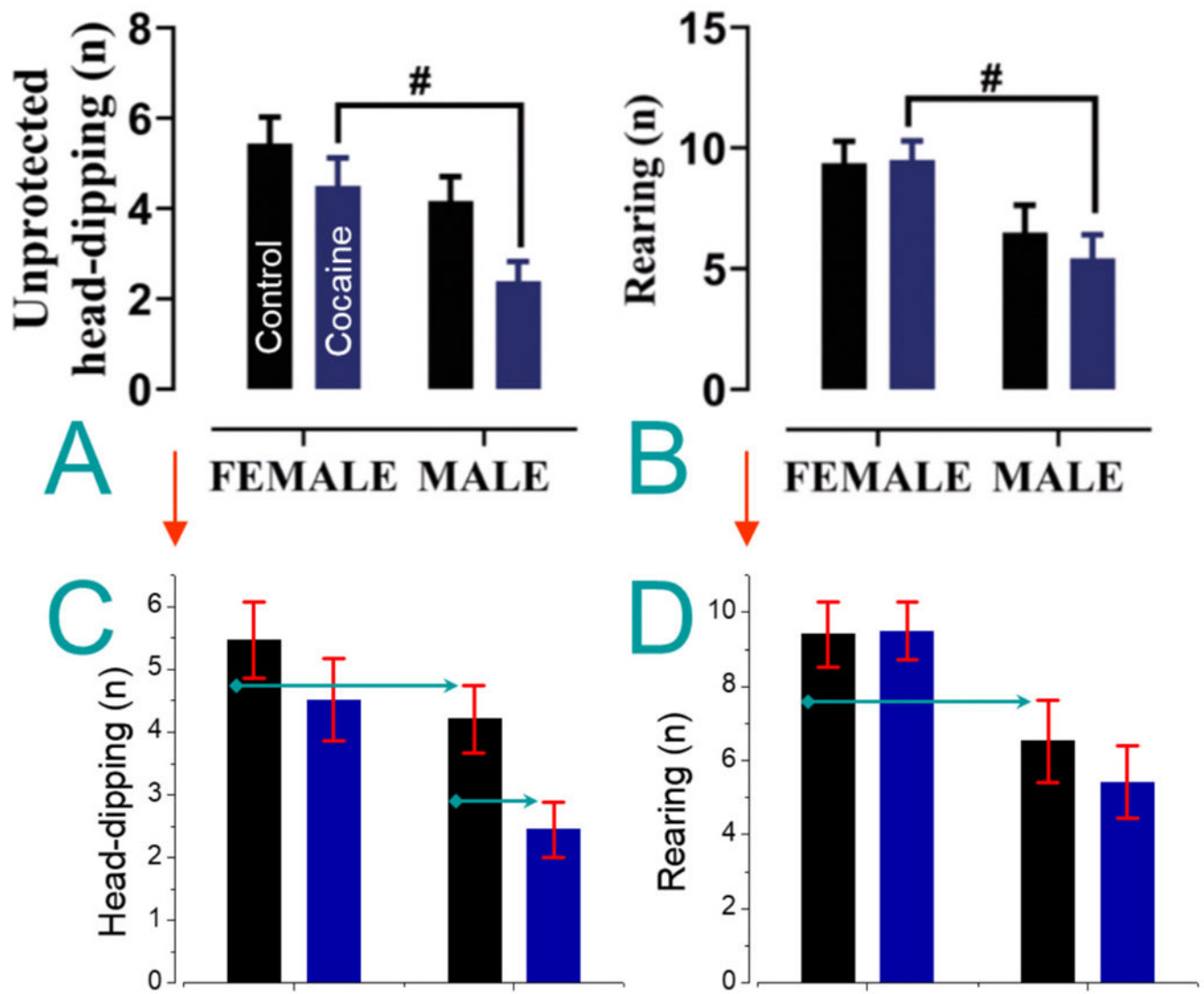
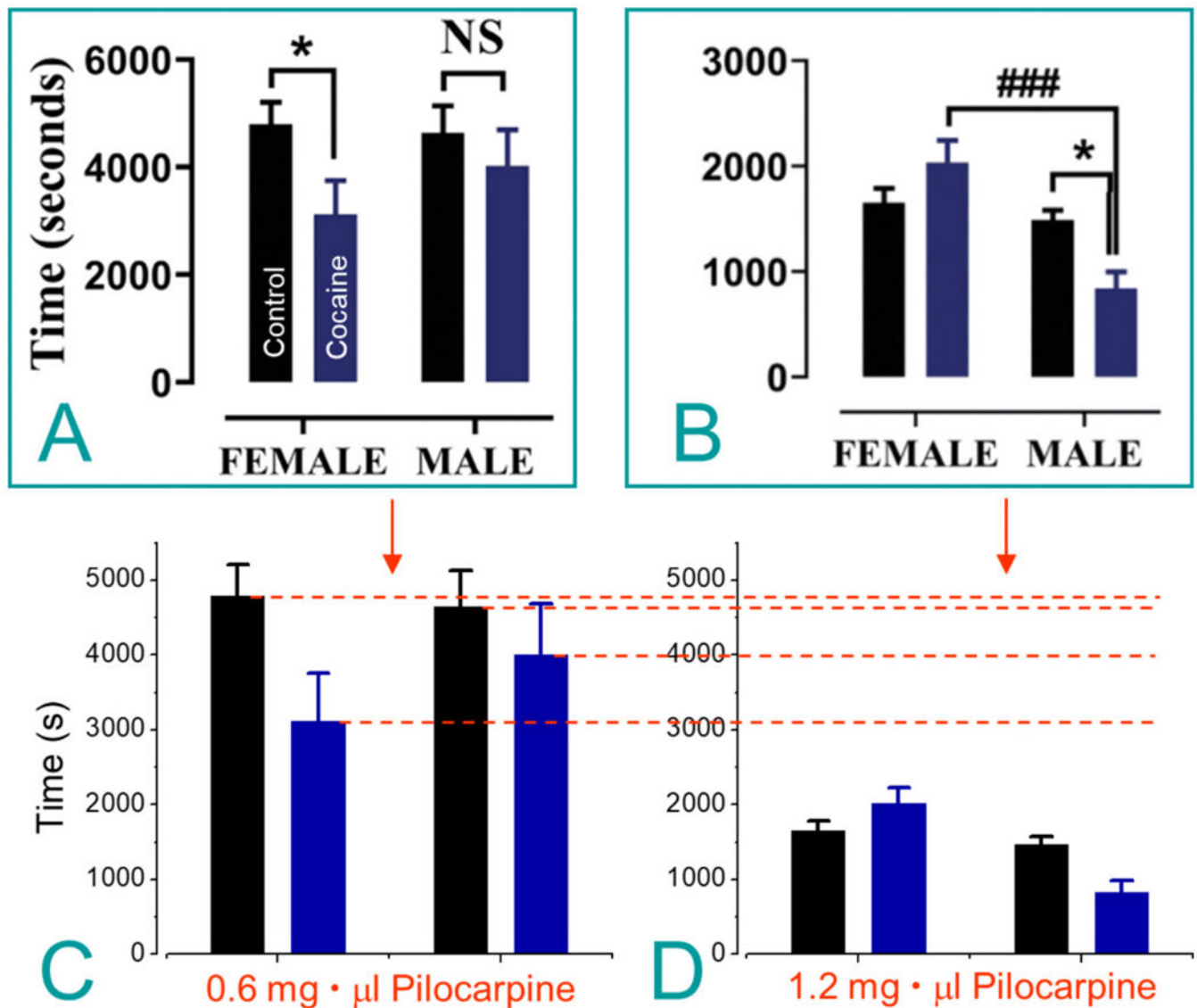
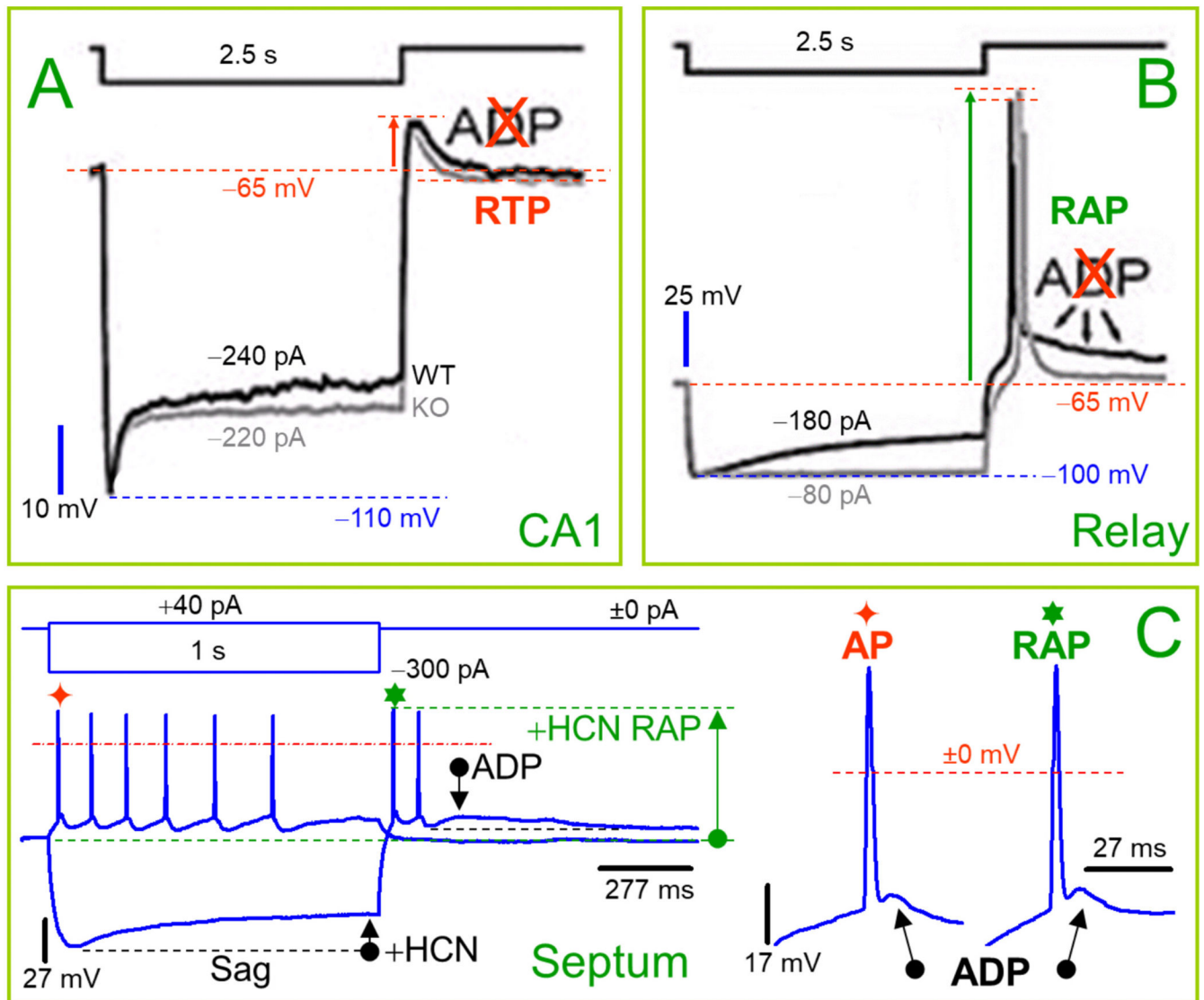


Fig. 15.

Anxiety-like behavior manifestation. (A) Comparisons among and between male and female rats (black) and those exposed to crack cocaine (blue). For comparison of frequency of unprotected head-dipping, an elevated plus-maze was used. (B) A similar tendency was concluded based on the frequency of rearing. (C) The same data in A are digitized (Fedorov, 2002). Note that ~2-fold decrease in crack cocaine-exposed males was not considered significant (arrows guide to learn non-overlapping SEM). Statistics are not always reliable, as values between males and females are different in control groups. (D) Why should the frequency of rearing between males and females as presented not be significant? Modified and highlighted (Pacheco et al., 2021).

**Fig. 16.**

Latency to onset of seizure. (A) Time span before SE observation after injection of 0.6 mg $\cdot$ μ l pilocarpine into the hippocampus. Latency was decreased in crack cocaine-exposed females but not males. (B) The opposite was true when the dose of pilocarpine was increased to 1.2 mg $\cdot$ μ l. (C and D) The same data from A and B are digitized (Fedorov, 2002). Contrary to assumption, a low dose of pilocarpine induces SE also in naïve controls, and the susceptibility is increased accordingly at 1.2 mg $\cdot$ μ l. Since the dose of crack cocaine is unknown, the effects of pilocarpine are unreliable, i.e., there is no internal control. Modified and highlighted (Pacheco et al., 2021).



HCN contribution to membrane potential under near-physiological conditions. (A) Patch-clamp recordings in pyramidal neurons of the CA1 region of hippocampus slices derived from wild-type (HCN2^{+/+}) and knockout (HCN2^{-/-}) mice in black and gray, respectively. The RMP is unknown, as the MP for comparison convenience is held at ‘around -65 mV by current injection’. Modified and highlighted (Ludwig et al., 2003). (B) Similar approach with thalamocortical relay neurons. Comparisons are inadequate since the hyperpolarization was achieved by currents of significantly different magnitudes at -180 vs. -80 pA for HCN2^{+/+} and HCN2^{-/-} mice. Despite the lower current amplitude, the MP response is saturated for HCN2^{-/-}. Also, the duration of steps (apparently 2.5 s) is too long, and ordinarily, neurons cannot cope with it. The rebound activities are mediated by HCN but not ADP. These are RTP - rebound tail potential (A) and RAP - rebound action potential (B). Both cases prove that the role of HCN2 is marginal for sag in these cells. (C) When and how should occur the ADP. The waveforms of MP during depolarization by +40 pA

and hyperpolarization by -300 pA current injections in rat LS neuron. The red dashed lines indicate the ± 0 mV levels where the overshoot initiates. The phenotype of both AP and rebound action potential (RAP). The ADP during the train is predominantly observed with regard to the 1st AP. Note the additional slower ADP after two RAPs. Scale bars are indicated.

Modified *figure 3* (Kodirov et al., 2014).

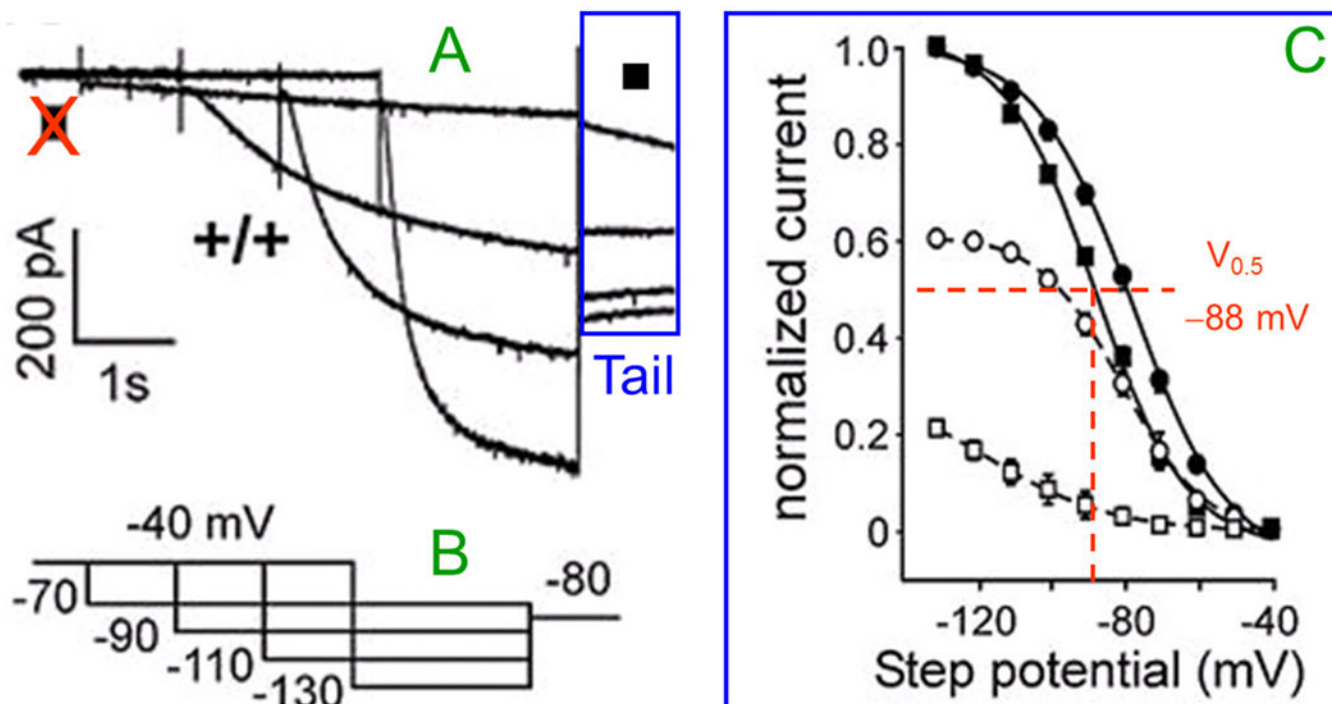


Fig. 18.

HCN current activation. Patch-clamp recording in thalamocortical relay neurons in a slice derived from a wild-type (HCN2^{+/+}) mouse. Normalized values are not from activation but from deactivation of channel, i.e., tail currents at a constant -80 mV. The latter reveals half-maximal activation voltage ($V_{0.5}$) of -88 mV (■). That of knockout -HCN2^{-/-} mice for relay neurons (□) cannot be determined from the graph, as normalization occurred with respect to WT. The same applies for CA1 pyramidal neurons of HCN2^{+/+} and HCN2^{-/-} mice (○ and ●).

Modified and highlighted (Ludwig et al., 2003).

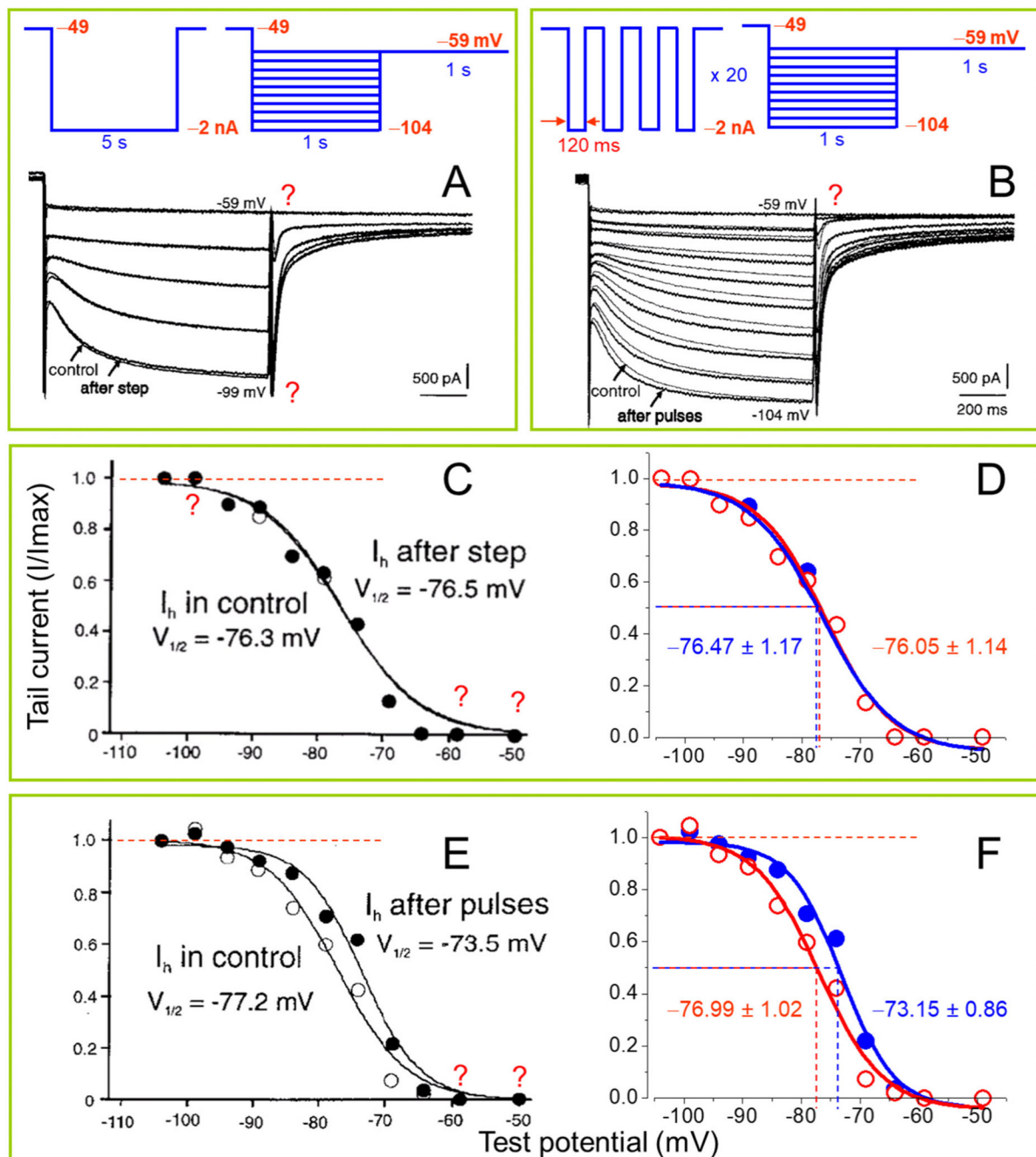


Fig. 19.

Bigger tail currents in thalamocortical cells of adult ferrets. (A) Priming with a -2 nA $\cdot$ 5 s step in current-clamp mode does not influence the subsequently recorded HCN currents in thalamocortical sagittal slices under voltage-clamp conditions. One cannot apply 2 nA currents to neurons in whole-cell patch-clamp recordings in contrast to electrophysiology with sharp micropipettes. If 'electrode resistances ranged between 80–120 M Ω for current-clamp recording and 50–90 M Ω for voltage-clamp recordings' it is confusing since both should be performed in the same neuron. (B) HCN channels were activated immediately

after $-2 \text{ nA} \cdot 20$ pulses at 4 Hz. For this purpose, the membrane potential was kept at -49 mV in current-clamp mode. However, the assumption that this could prevent the activation of I_h is incorrect since the extent of hyperpolarization by 2 nA pulses is excessive. After 120 ms of stimulation, the amplitude of I_h is bigger than the 'control', which simply means that a recovery occurred during up to 15 s. Thus, the design of the experiments is also inadequate. The brain most probably does not experience such periods of recurrent hyperpolarization. Besides, no one knows if the total durations, i.e., 120 ms vs. 5 s, are adequate for proper comparisons. Nonetheless, the important observations are bigger tails compared to other neurons in the CNS. (C) Values derived from tails in A. However, traces for the -104 mV step are missing (?) in A (see B). Besides, at -59 mV , there are no HCN or tail currents; therefore, including a data point of absolute 0 pA is incorrect. Interestingly, there is an additional outsider at either -49 or -50 mV (?). Therefore, the Boltzmann fit is slightly compromised. (E) Values derived from tails in B. The same concerns apply to data points (?). (D and F) The same data from C and E are digitized (Fedorov, 2002). Also, it is unclear why, at -99 mV , the ratio should slightly increase. It is possible that because of their faster tails, they are contaminated by capacitance currents. Modified and highlighted *figure 4* (Lüthi and McCormick, 1998).

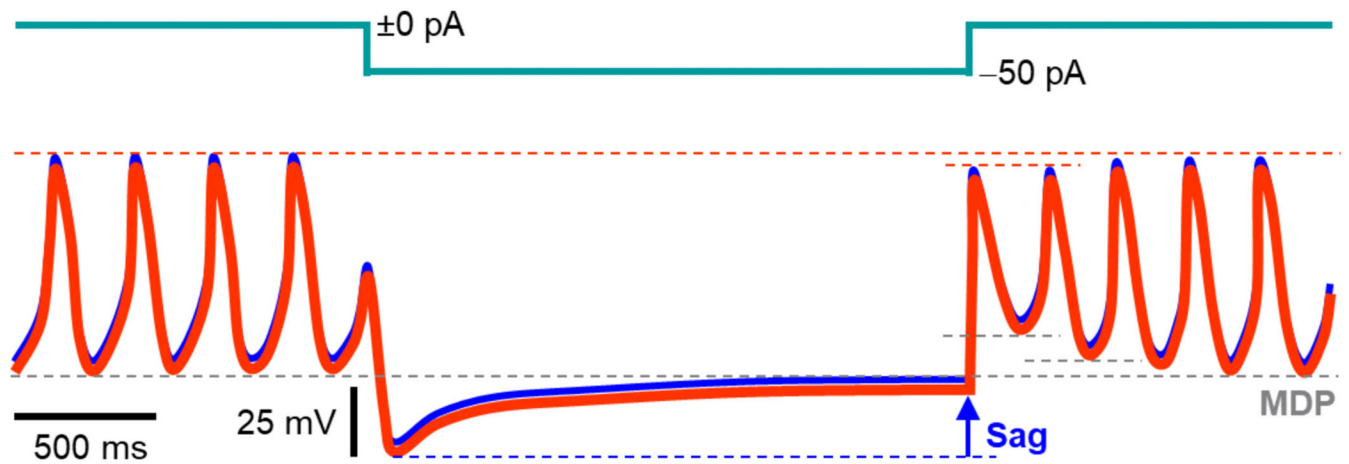


Fig. 20.

Spontaneous action potentials in rabbit S-A node. The timing and parameters of AP and MP coincided during two simultaneous recordings, which was due to the smaller dimension of the isolated tissue at 0.3×0.3 mm. During pacemaker activity, the MP was hyperpolarized by the injection of a small current. After peak hyperpolarization, the MP gradually returns to the baseline - MDP. This behavior is known as sag and reflects the activation of HCN. Note that the termination of the step has a small influence on peak AP amplitudes while having greater effects on MDP. This is also reflective of prior HCN activation, which leads to temporary rebound responses. Recording sites are 50 to 100 μm apart from the current injecting electrode for hyperpolarization, and the temperature of the extracellular solution was set to 35 $^{\circ}\text{C}$.

Highlighted, modified, and redrawn *figure 2* (Noma and Irisawa, 1976).

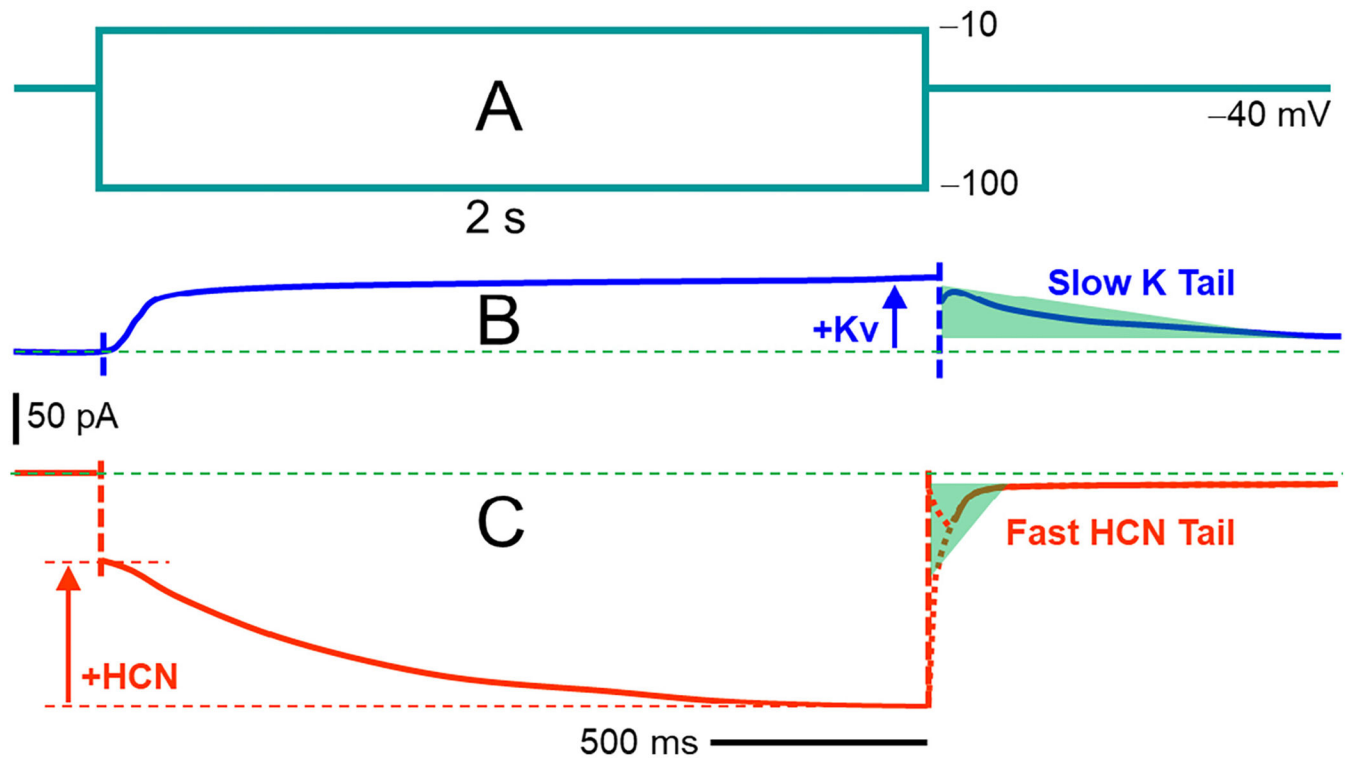


Fig. 21.

K and HCN currents in the S-A node. (A) Pulse protocol to activate inward and outward currents. The duration is retrospectively calculated from the time scale. (B) Simultaneous occurrences and sequentially activated outward K (blue) and inward HCN (red) currents in the same rabbit SAN tissue at 35 °C. Note that recordings were not conducted by the patch-clamp method, but qualities were comparable. The holding potential is identical at -40 mV for both currents. Even at low 30 mV depolarization, the activation of K current is fast, while that of HCN upon hyperpolarization by -60 mV is slow and gradually starts to saturate only at the end of the applied pulse. These features are also reflected by deactivation - tails, albeit slower and faster time kinetics, respectively. The onset of HCN tail may follow two theoretical paths, as indicated by dashed lines after capacitive currents (interpretation is based on established literature, Kodirov). In both cases, the currents do not completely return to baseline during ~1 s after the termination of steps.

Highlighted, modified, and redrawn for clarity, *figure 9* (Noma and Irisawa, 1976).

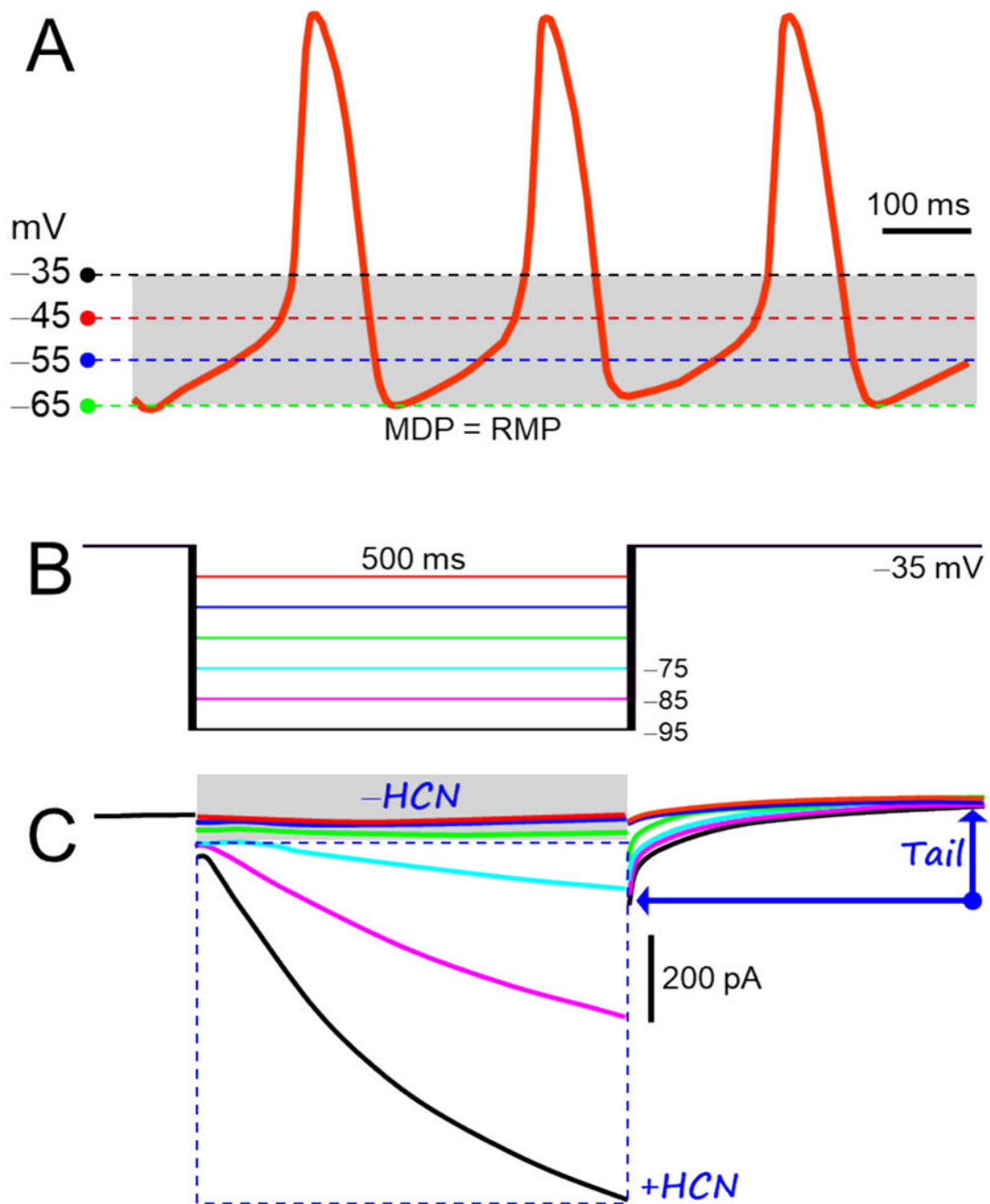


Fig. 22.

The absence of funny HCN current within the pacemaking window. (A) Intrinsic excitability of SAN myocyte. (B) From A derived pulse protocol for recording I_Q in the same cell. (C) Typical current phenotype upon stronger hyperpolarization. Note the insignificant current even at -65 mV, corresponding to the MDP of this cell. Channel deactivation - tail current is observed only after adequate and significant activation (dashed blue box, +HCN). Pacemaker range and current responses in A and C are gray-shaded, while each step and resulting inward I_Q in B and C are color-matched. For clarity, all traces are redrawn.

Modified (Denyer and Brown, 1990).

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